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A simple realized factor-based portfolio: improving minimum variance portfolio performance by incorporating low-frequency betas

WANBO LU[†] and YIFU WANG ^{*}

[†]School of Management Science and Engineering, Southwestern University of Finance and Economics, Chengdu, People's Republic of China

[‡]School of Statistics and Data Science, Southwestern University of Finance and Economics, Chengdu, People's Republic of China

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Accurate estimation of large covariance and precision matrices is an essential prerequisite of portfolio selection and financial risk management. Among factor-based covariance estimators, high-frequency data provides additional information but also introduces noise, which could degrade the estimation of betas. In this paper, we propose a new Mixed-frequency and Factor-based (MFACE) combining high-frequency (intraday) data and low-frequency (daily) betas. We apply approximate factor structure to construct the high-dimensional minimum variance portfolio. We establish the consistency and obtain the convergence rate of the covariance estimator and the corresponding precision matrix estimator. A comprehensive simulation study investigates the estimation accuracy under different combinations of intraday sampling frequency, intraday sample size and daily sample size. Out-of-sample forecasts demonstrate that our estimator explains the daily volatility of the equally-weighted portfolio pretty well. The minimum variance portfolio that embodies low-frequency betas also achieves the minimum risk at relatively low cost among several estimators.

Keywords: Mixed-frequency data; Approximate factor structure; Minimum variance portfolio; Location-based thresholding; Microstructure noise

JEL Classifications: C51, G11

1. Introduction

In the realm of portfolio management, the quest for optimal risk-adjusted returns has led to the development of various strategies and methodologies. Among them, the mean-variance portfolio (Markowitz 1952) stands out as a cornerstone approach, aiming to construct a portfolio with the lowest possible volatility while maintaining a desirable level of return. Two crucial inputs are estimates of the expected returns and covariance matrices. However, estimating expected returns is more difficult than the estimation of covariance structure (Merton 1980, Jagannathan and Ma 2003, Petukhina *et al.* 2024). In this context, the minimum variance portfolio (MVP) policy is a promising alternative to circumvent the issue. In contrast to the traditional Markowitz approach, MVP parameterizes portfolio weights as a function of the inverse covariance matrix of stock returns. The policy

does not require estimating expected stock returns as in the traditional Markowitz approach (Cai *et al.* 2020, Petukhina *et al.* 2024). The estimation of covariance matrices is crucial for portfolio allocation and risk management in financial markets.

The advancement of computer technology has spurred the integration of intraday high-frequency data modeling into covariance estimation and portfolio construction. The utilization of high-frequency financial data significantly enhances the available sample size for volatility estimation, thereby shortening the time window and improving adaptability to local covariations (Fan *et al.* 2012). While high-frequency data enriches covariance estimation with more information, the estimation with large number of assets still poses challenges.

Primarily, estimating volatility in large dimensions encounters the abundance of parameters. Consequently, prediction accuracy may suffer, which is crucial in constructing optimal portfolios (Chan *et al.* 1999, Jagannathan and Ma 2003, Fan

* Corresponding author. Email: yifu_wang@outlook.com

et al. 2008). Scholars have started to address this by modeling the covariance of financial assets using both high-frequency data high-dimensional techniques. Wang and Zou (2010) explored the asymptotic property of vast integrated covariance matrices under the sparsity assumption. Fan *et al.* (2012) proposed the use of ‘pairwise-refresh time’ and ‘all-refresh time’ methods for the estimation of large integrated covariance matrix. Other studies within this framework include Tao *et al.* (2013a) and Tao *et al.* (2013b).

Also, with the increase of sampling frequency, covariance estimation is largely influenced by market microstructure effects, such as bid-ask spreads, price discontinuities and asynchronous transactions (Epps 1979, Lo and MacKinlay 1990). To address this issue, Zhang *et al.* (2005), Bandi and Russell (2008) and Hansen and Lunde (2006) propose various nonparametric estimators. Kim *et al.* (2022) and Chen *et al.* (2024) develop realized beta estimators using high-frequency financial data contaminated by microstructure noises. However, these improvements come at the cost of increased computational complexity.

In this paper, we introduce a novel approach to portfolio construction that addresses this limitation by incorporating low-frequency daily betas into the MVP framework. Our proposed method leverages the benefits of both high-frequency and low-frequency data, thereby offering enhanced risk-adjusted returns. Also, the low-frequency betas mitigate the bias arising from the market microstructure effect, leading to superior performance compared to the estimators based on low-frequency and high-frequency factor models and popular realized covariance estimators.

Factor structures have been found to be useful to tackle the ‘curse of dimensionality’. In case of a large number of assets, factor models can significantly reduce the parameters in covariance estimation (De Nard *et al.* 2021). Fan *et al.* (2016) constructed a high-frequency covariance estimator based on the approximate factor model (HFFBC), after sorting the assets by their Global Industry Classification Standard (GICS) codes. On this foundation, some scholars applied factor structure to covariance matrix and constructed optimal portfolio. Ait-Sahalia and Xiu (2017), Pelger (2019) and Kim *et al.* (2018) estimated covariance based on the principal component (PCA) method. Bollerslev *et al.* (2019) estimated the realized covariance matrices based on the assumption of a factor structure for the noise component. The existing literatures mainly focus on nonparametric techniques considering the effect of market microstructure noise. The advantage of imposing factor structure over regularization methods (Bickel and Levina 2008) is that it can guarantee positive (semi-)definiteness of the resulting covariance estimators. Well-conditioned covariance estimates are important for the construction of MVP, since it requires the estimation of precision matrices (inverse covariance matrices).

In addition to high-frequency data, we consider the information provided by low-frequency data to relieve the market microstructure effects. Specifically, we apply approximate factor structure to high-dimensional asset returns and propose a Mixed-frequency and FActor-based Covariance Estimator (MFACE). The MFACE employs high-frequency (intraday) data to estimate factor covariances and idiosyncratic

covariances while the factor loadings are obtained using low-frequency (daily) data. This paper makes useful supplements in that we extend the information set to accommodate the factor structure in both high-frequency and low-frequency data to alleviate the impact of market microstructure noise.

Our paper is connected to recent and growing branch of the literature that explores the potential of mixed-frequency data in covariance estimation (Tao *et al.* 2011, Bannouh *et al.* 2012, Halbleib and Voev 2016, Peng *et al.* 2022). Tao *et al.* (2011) propose a methodology that combines high-frequency volatility matrix estimation with low-frequency dynamic models to infer large volatility matrices accurately. Halbleib and Voev (2016) introduce a mixed forecasting approach that combines forecasts derived from different information sets to predict large-dimensional covariance matrices. Bannouh *et al.* (2012) introduce the Mixed-Frequency Factor Model (MFFM) as a novel approach to estimate vast-dimensional covariance matrices of asset returns. Peng *et al.* (2022) propose a Mixed-Frequency Dynamic Factor Model (MFDFM) that incorporates both high-frequency and low-frequency variation in factor loadings. Compared to these studies, we employ approximate factor model to allow more flexible residual covariance structure.

The key contribution of our approach lies in the utilization of low-frequency daily betas, which capture the long-term relationships between asset returns and market factors. By integrating these low-frequency betas into the portfolio optimization process, we aim to mitigate the adverse effects of market microstructure, such as bid-ask spreads and price impact, which are often prevalent in high-frequency data.

Moreover, our realized factor-based portfolio model offers a simple yet effective solution that is easy to implement for scholars and practitioners alike. By combining the predictive power of low-frequency betas with the precision of high-frequency data, our approach provides investors with a robust framework for portfolio construction that can adapt to changing market conditions and deliver superior risk-adjusted returns over time.

The rest of the paper is organized as follows. Section 2 sets up the model and constructs the estimators. In section 3, we prove that the estimator with approximate factor structure holds consistency in high-dimensional setting. Section 4 demonstrates that the estimator can mitigate the influence of market microstructure effects. In section 5, by comparing the performance of the minimum variance portfolio constructed by alternative estimators, ‘mixed-frequency low rank plus sparsity’ structure can effectively raise the performance of the minimum variance portfolio with lower transaction cost. We conclude in section 6.

2. Approximate factor model and mixed-frequency covariance estimator

2.1. Mixed-frequency factor model

Assume that we observe a large intraday panel of assets and factor prices, Y and F at $0, \Delta_n, 2\Delta_n, \dots, 1, 1 + \Delta_n, \dots, t$, where Δ_n is the sampling frequency. We denote the factor loading matrix that projects Y on F as β and have the

following continuous-time factor model,

$$Y_t = \beta F_t + Z_t, \quad (1)$$

where Y is a d -dimensional vector process, F is an r -dimensional observable factor process, Z is the idiosyncratic component. It's worth noticing that the covariance structure of Z does not have to be diagonal, which means the model is actually the so called approximate factor model proposed by Chamberlain and Rothschild (1983), see Assumption 2 for details. We estimate β from low-frequency data in practice and identify β from the following low-frequency factor model,

$$Y_\tau = \mathbf{B}F_\tau + Z_\tau. \quad (2)$$

Let F_{it} , Y_{it} , and Z_{it} be the i th elements of Y_t , F_t , and Z_t , and $E(Z_\tau|F_\tau) = 0$. Note that $\tau = t - T + 1, t - T + 2, \dots, t$, represents integer times within the high-frequency timestamps, and the interval lengths t and T corresponding to high-frequency and low-frequency time series are generally different. We further assume that $\{F_\tau\}$ is stationary and ergodic to identify the factor loadings.

2.2. Basic assumptions and mixed-frequency covariance estimator

To complete the mixed-frequency specification, we need the following assumptions.

ASSUMPTION 1 (High-frequency Factor Structure)

(a) F_t and Z_t are $It\hat{o}$ semi-martingales, that is,

$$\begin{aligned} F_t &= \int_0^t h_s ds + \int_0^t \eta_s dW_s, \\ Z_t &= \int_0^t f_s ds + \int_0^t \gamma_s dB_s. \end{aligned} \quad (3)$$

We denote the spot covariance of F_t and Z_t as $e_t = \eta_t \eta_t^\top$, $g_t = \gamma_t \gamma_t^\top$. W_t and B_t are independent Brownian motions. In addition, h_t and f_t are progressively measurable. Finally, the processes η_t and γ_t are càdlàg, and e_t , e_{t-} , g_t , and g_{t-} are positive-definite, e_{t-} and g_{t-} are the left limits of the corresponding processes, respectively. Moreover, for all $1 \leq i, j \leq r$, $1 \leq k, l \leq d$, $|\mathbf{B}_{kj}| \leq K$, for some $K > 0$, there exists a locally bounded process h_t , such that $|h_{i,s}|$, $|\eta_{ij,s}|$, $|\gamma_{kl,s}|$, $|e_{ij,s}|$, $|f_{kl,s}|$ and $|g_{kl,s}|$ are all bounded by H_s for $0 \leq s \leq t$.

(b) For any $1 \leq j \leq r$, $1 \leq k \leq d$, we have $[Z_{k,s}, F_{j,s}] = 0$, for any $0 \leq s \leq t$, where $[\cdot, \cdot]$ denotes the quadratic covariation.

Assumption 1(b) is similar in spirit to the usual exogeneity condition in a standard regression setting, but it is tailored to this asymptotic framework. This assumption only imposes a restriction on the sample paths of F and Z , which does not rule out any long-term dependence between these two processes.

Assumption 1 imposes a factor structure on the covariance matrix of Y , the spot covariance of Y_s can be written as

$$c_s = \mathbf{B}e_s\mathbf{B}^\top + g_s, \quad 0 \leq s \leq t. \quad (4)$$

Writing $\Sigma^{\text{HFFBC}} = \frac{1}{t} \int_0^t c_s ds$, $\Gamma = \frac{1}{t} \int_0^t g_s ds$, $E = \frac{1}{t} \int_0^t e_s ds$, we obtain

$$\Sigma^{\text{HFFBC}} = \beta E \beta^\top + \Gamma. \quad (5)$$

Thus, the matrix form of mixed-frequency covariance decomposition can be denoted as

$$\Sigma := \mathbf{B}E\mathbf{B}^\top + \Gamma. \quad (6)$$

ASSUMPTION 2 (Sparsity of Γ) Γ is a block diagonal matrix with eigenvalues bounded from above and below by some constants almost surely. The set of its nonzero entries, S , is known prior to the estimation.

Assumption 2 balances theoretical application against practical application, which imposes block diagonal structure. Our assumption actually falls into the approximate factor model. The approximate factor model is more general than classical factor model in that it takes into account the non-diagonal covariance elements of the idiosyncratic components. We will provide an economically sensible approach that takes advantage of the GICS code on the underlying stocks and proceed to empirically validate this assumption.

We implement a plug-in type estimator for (6) in the mixed-frequency setting. First we stack the processes Y and F into $U = (Y^\top, F^\top)^\top$. Then we have

$$\begin{aligned} \Pi &:= \frac{1}{t} \int_0^t [dU_s, dU_s] ds = \frac{1}{t} \int_0^t \begin{pmatrix} \beta e_s \beta^\top + g_s \beta e_s \\ e_s \beta^\top e_s \end{pmatrix} ds \\ &=: \begin{pmatrix} \Pi^{11} & \Pi^{12} \\ \Pi^{21} & \Pi^{22} \end{pmatrix}. \end{aligned} \quad (7)$$

With this representation, our estimator has a simple form. We apply the realized covariance estimator to the process:

$$\begin{aligned} \hat{\Pi} &= \frac{1}{t} \sum_{j=1}^{\lfloor t/\Delta_n \rfloor} (\Delta_j^n U)(\Delta_j^n U)^\top, \\ \text{where } \Delta_j^n U &= U_{j\Delta_n} - U_{(j-1)\Delta_n}. \end{aligned} \quad (8)$$

Once $\hat{\Pi}$ is calculated, we can further construct estimators for $\hat{E}$ and $\hat{\Gamma}$:

$$\hat{E} = \hat{\Pi}^{22}, \quad (9)$$

$$\hat{\Gamma} = \hat{\Pi}^{11} - \hat{\Pi}^{12}(\hat{\Pi}^{22})^{-1}\hat{\Pi}^{21}. \quad (10)$$

Obviously, $\hat{\Gamma}$ is positive-semidefinite. The current estimator of $\hat{\Gamma}$ does not incorporate the additional information about the residual matrix Γ , that is, Assumption 2. To take this into account, we will allow for thresholding the residual covariance matrix. We will concentrate on hard thresholding and will denote a hard-thresholded matrix as

$$\hat{\Gamma}^S = (\hat{\Gamma}_{ij} 1_{(i,j) \in S}), \quad (11)$$

for some choice of an index set S .

In section 4, we will directly impose the block diagonal structure. How to choose the information set S will be discussed in section 5.

Under low-frequency setting, we can estimate $\mathbf{B}$ by the least square method when the factors are observable: $\widehat{\mathbf{B}} = (\widehat{\mathbf{b}}_1^\top, \dots, \widehat{\mathbf{b}}_d^\top)^\top$, where

$$\widehat{\mathbf{b}}_i = \arg \min_{\mathbf{b}_i} \frac{1}{Td} \sum_{\tau=T-t+1}^t \sum_{i=1}^d (Y_{i\tau} - \mathbf{b}_i^\top F_\tau)^2. \quad (12)$$

We can now define our mixed-frequency estimator for the covariance matrix:

$$\widehat{\Sigma}^{\text{MFACE}} = \widehat{\mathbf{B}}\widehat{\mathbf{E}}\widehat{\mathbf{B}}^\top + \widehat{\Gamma}^S. \quad (13)$$

Because $\widehat{\Gamma}$ is positive-semidefinite, each of its blocks on the diagonal is positive-semidefinite, which implies that $\widehat{\Gamma}^S$ is also positive-semidefinite. In this case, $\widehat{\Sigma}^{\text{MFACE}}$ is also positive-semidefinite, since $\widehat{\mathbf{B}}\widehat{\mathbf{E}}\widehat{\mathbf{B}}^\top$ is positive-semidefinite. This feature of our estimator is very appealing in some applications, for example, minimum variance portfolio.

2.3. How low-frequency betas mitigate the impact of microstructure noise

The notion that the observed transaction price in high-frequency financial data is the unobservable efficient price $\widetilde{Y}_t$ plus some noise component ϵ_t due to the imperfections of the trading process is a well established concept in the market microstructure literature (see for instance Black 1986), thus

$$Y_t = \widetilde{Y}_t + \epsilon_t.$$

In the simplest possible case, the ϵ_t s are i.i.d. noise with mean zero and constant variance and are independent of $\widetilde{Y}_t$. ϵ_t have rich dynamics induced by market frictions. According to Zhang *et al.* (2005), realized variance is a biased estimator of the integrated variance. It's fairly reasonable to assume that the bias are in the factor loadings in the observable factor model since the estimated factor loadings would not achieve unbiasedness under this noisy condition. We further decompose the factor loadings as follows:

$$\begin{aligned} Y_t &= (\text{true}(\beta) + \text{bias}(\beta))F_t + Z_t \\ &= \text{true}(\beta)F_t + \text{bias}(\beta)F_t + Z_t, \end{aligned}$$

where $\text{bias}(\beta)$ represents the bias in β induced by the market microstructure. As can be seen from the above equation, the impact of microstructure noise is in the term $\text{bias}(\beta)F_t$. We view $\mathbf{B}$ as a proxy for $\text{true}(\beta)$ since single-day betas based on high-frequency data are very noisy due to the non-synchronicity between factor returns and stock returns, as shown in figure 1. $\mathbf{B}$ will be estimated using daily data. The unobservable efficient price Y_t can be expressed as

$$Y_t = \mathbf{B}F_t + \widetilde{Z}_t,$$

where $\widetilde{Z}_t$ is approximately the sum of $\text{bias}(\beta)F_t$ and Z_t .

3. Asymptotic properties of the estimator

Throughout the article, we use $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ to denote the minimum and maximum eigenvalues of a matrix A . In addition, we use $\|A\|_1$, $\|A\|$ and $\|A\|_F$ to denote the $\mathbb{L}_1$ norm, the operator norm (or $\mathbb{L}_2$ norm), and the Frobenius norm of a matrix A , that is, $\max_j \sum_i |A_{ij}|$, $\sqrt{\lambda_{\max}(A^\top A)}$, and $\sqrt{\text{Tr}(A^\top A)}$, respectively. We also use $\|A\|_{\text{MAX}} = \max_{i,j} |A_{ij}|$ to denote the $\mathbb{L}_\infty$ norm of A on the vector space.

We impose the following assumptions for the derivation of the asymptotic properties.

ASSUMPTION 3 (Low-frequency Factor Structure)

- (a) Z_τ 's are independent and identically distributed (i.i.d.) with mean vector zero.
- (b) There exist $r_1 > 0$ and $b > 0$, such that for any $s > 0$ and $i \leq d$,

$$P(|Z_{i\tau}| > s) \leq \exp(-(s/b)^{r_1}).$$

- (c) There exist constants $C_1, C_2 > 0$ such that $C_1 \leq \lambda_{\min}(\text{cov}(F_\tau)) \leq C_2$.
- (d) There exist $r_2 > 0$ and $b_2 > 0$ such that for any $s > 0$ and $i \leq r$,

$$P(|F_{i\tau}| > s) \leq \exp(-(s/b_2)^{r_2}).$$

- (e) $\|d^{-1}\mathbf{B}^\top \mathbf{B} - \Omega\| = o(1)$ for some $r \times r$ symmetric positive definite matrix Ω such that $\lambda_{\min}(\Omega)$ and $\lambda_{\max}(\Omega)$ are bounded away from both zero and infinity.

Assumption 3 contains common conditions required to investigate the asymptotic properties in factor models. Condition (b) requires the distributions of $(Z_{1\tau}, \dots, Z_{d\tau})$ to have exponential-type tails. (c) is useful in the discussion of asymptotic properties of the least square estimator. (d) allows us to apply the Bernstein type inequality for weakly dependent data. (e) requires that the factors should impact every individual time series.

We further introduce the strong mixing conditions to conduct asymptotic analysis of the least square estimates. Let $\mathcal{F}_{-\infty}^0$ and $\mathcal{F}_T^\infty$ denote the σ -algebras generated by $\{(F_\tau, Z_\tau) : -\infty \leq \tau \leq 0\}$ and $\{(F_\tau, Z_\tau) : T \leq \tau \leq \infty\}$ respectively. In addition, define the mixing coefficient:

$$\alpha(T) = \sup_{A \in \mathcal{F}_{-\infty}^0, B \in \mathcal{F}_T^\infty} |P(A)P(B) - P(AB)|. \quad (14)$$

ASSUMPTION 4 There exist positive constants γ and C such that for all $\tau \in \mathbb{Z}^+$,

$$\alpha(\tau) \leq \exp(-C\tau^\gamma).$$

The strong mixing assumption in Assumption 4 enables us to apply the Bernstein's inequality in the technical proofs.

We consider the setting where the number of assets and factors increase, as the sampling frequency shrinks to zero, that is, $d, r \rightarrow \infty$ and $\Delta_n \rightarrow 0$. We allow the number of factors r to grow at a lower rate than the dimensionality of assets d , denoted as $r = o(d)$, which is quite reasonable in the factor model.

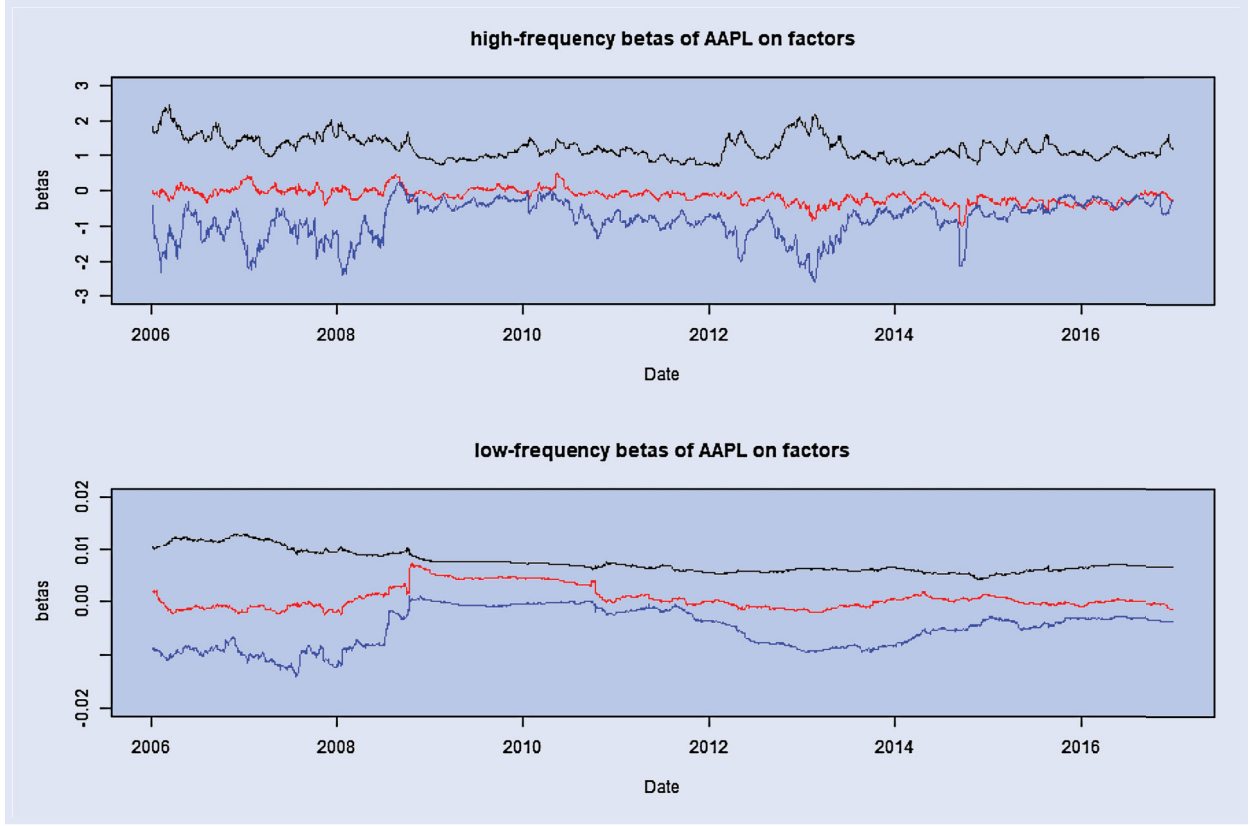


Figure 1. Betas of AAPL to MKT, redSMB and blueHML in high-frequency factor model and low-frequency factor model.

ASSUMPTION 5 (a) $r\sqrt{\Delta_n \log d} = o(1)$.
 (b) $r^4 \log d/T = o(1)$.

We can infer from condition (a) that the sampling frequency Δ_n has to be sufficiently high to accommodate the increasing dimensionality. Condition (b) indicates that the number of low-frequency observations should be large enough to compensate the loss in efficiency due to increasing dimensionality. Note that no restrictions are imposed on the relationship between Δ_n and $1/T$, delivering flexibility in numerical applications.

We need the following lemmas to derive the consistency of MFACE and its convergence rate.

LEMMA 1 Under Assumption 3, we have $\max_{i \leq d} \|\hat{\mathbf{b}}_i - \mathbf{b}_i\| = O_p(\sqrt{r \log d/T})$, $\|\hat{\mathbf{B}} - \mathbf{B}\|_F^2 = O_p(dr/T)$.

LEMMA 2 Under Assumptions 1, 2, 4 and $m_d r \sqrt{\Delta_n \log d} = o(1)$, we have

- (a) $\mathbb{P}(\|\hat{\mathbf{E}} - \mathbf{E}\|_{\text{MAX}} > C_0 \sqrt{\Delta_n \log d}) = O(r^2 d^{-3})$.
- (b) $\mathbb{P}(\|\hat{\mathbf{E}} - \mathbf{E}\|_F > C_0 r \sqrt{\Delta_n \log d}) = O(r^2 d^{-3})$.
- (c) $\mathbb{P}(\max_{1 \leq j \leq r, 1 \leq l \leq d} |\sum_{i=1}^{\lfloor T/\Delta_n \rfloor} (\Delta_i^n F_j)(\Delta_i^n Z_l)| > C_0 \sqrt{\Delta_n \log d}) = O(rd^{-2})$.
- (d) $\mathbb{P}(\max_{1 \leq l, k \leq d} |\hat{\Gamma}_{lk} - \Gamma_{lk}| \geq C_0 r \sqrt{\Delta_n \log d}) = O(d^{-1})$.
- (e) $\mathbb{P}(\|\hat{\Gamma}^S)^{-1} - \Gamma^{-1}\| > C_0 m_d r \sqrt{\Delta_n \log d}) = O(d^{-1})$.
- (f) $\mathbb{P}(\lambda_{\min}(\hat{\Gamma}^S) \geq \frac{1}{2} \lambda_{\min}(\Gamma)) > 1 - O(d^{-1})$.

Lemma 1(a), (b) are LEMMA B.1. (i) and LEMMA B.3. (ii) from Fan *et al.* (2011), Lemma 2(a), (b), (c), (d), (e), (f) are LEMMA A.2. (i), (ii), (iii), (viii) and LEMMA A.3.

(iii), (ii) from Fan *et al.* (2016), respectively. We present the convergence rate of our MFACE estimator in Theorem 1.

THEOREM 1 Under Assumptions 1–5, we have

$$\|\hat{\Sigma}^{\text{MFACE}} - \Sigma\|_{\text{MAX}} = O_p\left(\sqrt{r^4 \log d/T} + \sqrt{r^2 \Delta_n \log d}\right). \quad (15)$$

Proof The proof of this theorem is in Appendix 1. ■

We are interested in the consistency of $\hat{\Sigma}^{\text{MFACE}}$ under the elementwise norm, since it's particularly relevant to the risk of solutions to the portfolio selection problem:

$$\min_w w^\top \hat{\Sigma}^{\text{MFACE}} w, \quad \text{subject to } w^\top \mathbf{1} = 1, \quad \|w\|_1 \leq \gamma. \quad (16)$$

When γ is small and binding, the optimal portfolio is sparse, that is, the dimension of the portfolio is not actually so high and many weights are zero. We need the following Lemma 3 to develop asymptotic theory for the precision matrix estimator.

LEMMA 3 Under Assumptions 1–5, and $m_d r \sqrt{\Delta_n \log d} = o(1)$, we have

- (a) $\mathbb{P}(\|\hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}} - \mathbf{B}^\top \Gamma^{-1} \mathbf{B}\| > C_0 m_d r \sqrt{\Delta_n d^2 \log d}) = O(d^{-1})$.
- (b) $\mathbb{P}(\|(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}}) - (\mathbf{E}^{-1} + \mathbf{B}^\top \Gamma^{-1} \mathbf{B})\| > C_0 m_d r \sqrt{\Delta_n d^2 \log d}) = O(d^{-1})$.
- (c) $\mathbb{P}(\|\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}}\| > C_0 d^{-1}) = O(d^{-1})$.

- (d) $\mathbb{P}(\|\widehat{\mathbf{B}}(\widehat{\mathbf{E}}^{-1} + \widehat{\mathbf{B}}^\top(\widehat{\Gamma}^S)^{-1}\widehat{\mathbf{B}})^{-1}\widehat{\mathbf{B}}^\top(\widehat{\Gamma}^S)^{-1}\| > C_0) = O(d^{-1}).$
(e) $\mathbb{P}(\|\widehat{\mathbf{B}}(\widehat{\mathbf{E}}^{-1} + \widehat{\mathbf{B}}^\top(\widehat{\Gamma}^S)^{-1}\widehat{\mathbf{B}})^{-1}\widehat{\mathbf{B}}^\top\Gamma^{-1}\| > C_0) = O(d^{-1}).$

Proof The proof is in Appendix 2. ■

We define the degree of sparsity $m_d = \max_{1 \leq i \leq d} \sum_{1 \leq j \leq d} 1_{\{\Gamma_{ij} \neq 0\}}$ and present the consistency of the precision matrix estimator in the following theorem.

THEOREM 2 Suppose that Assumptions 1–5 hold, and $m_d r \sqrt{\Delta_n \log d} = o(1)$, then we have

$$\|(\widehat{\Sigma}^{\text{MFACE}})^{-1} - \Sigma^{-1}\| = O_p\left(m_d r \sqrt{\Delta_n \log d} + \sqrt{\frac{r}{T}}\right). \quad (17)$$

Proof The proof of this theorem is in Appendix 3. ■

The convergence rate of the precision matrix under the operator norm depends on both the sampling frequency of high-frequency part and the length of low-frequency data. When the high-frequency part dominates in the right-hand side of (17), the degree of sparsity and the dimensions of both returns and common factors will notably affect the theoretical results. When the low-frequency part dominates, the dimension of common factors will mainly count. In our framework, Γ is sparse whereas Σ is not, since our factors are pervasive.

4. Simulation study

In this section, we examine the performance of our estimator in the presence of both asynchronous trading and microstructure noise.

4.1. Simulation setup

To compare with HFFBC proposed by Fan *et al.* (2016), we first simulate the high-frequency part from a continuous-time r -factor model of d assets specified as:

$$\begin{aligned} dY_{i,t} &= \sum_{j=1}^r \beta_{ij} dF_{j,t} + dZ_{i,t}, \\ dF_{j,t} &= b_j dt + \sigma_{j,t} dW_{j,t}, \quad dZ_{i,t} = \gamma_i^\top dB_{i,t}, \end{aligned}$$

where W_j is a standard Brownian motion and B_i is a d -dimensional Brownian motion, for $i = 1, 2, \dots, d$, and $j = 1, 2, \dots, r$. They are mutually independent. F_j is the j th observable factor. The covariance matrix of Z is a block diagonal matrix, denoted by Γ , so that $\Gamma_{il} = \gamma_i^\top \gamma_l$. We allow for time-varying $\sigma_{j,t}$ which evolves according to the following system of equations:

$$d\sigma_{j,t}^2 = \kappa_j(\theta_j - \sigma_{j,t}^2)dt + \eta_j \sigma_{j,t} d\widetilde{W}_{j,t}, \quad j = 1, 2, \dots, r,$$

where $\widetilde{W}_j$ is a standard Brownian motion with $\mathbb{E}[dW_{j,t} d\widetilde{W}_{j,t}] = \rho_j dt$.

We first simulate all stocks at 1-s frequency, then contaminate and censor the data to simulate the effects of microstructure noise and asynchronous trading. As is common, we add normal random noise with mean zero and variance σ_{MN}^2 to the simulated log prices before censoring. To censor the data, we first generate the desired number of observed prices for each of the d assets. These values are drawn from a truncated log-normal distribution so that the number of daily observations match the empirical pattern. The log-normal distribution has parameters $\mu = 2,500$ and $\sigma = 0.8$. The lower and upper truncation boundaries are 500 and 23,400, respectively. Finally, we subsample the selected number of observations within a day according to a uniform distribution. Our procedure results in a panel of transaction prices, which are contaminated by microstructure noise, and which appear to arrive according to Poisson processes.

To apply our mixed-frequency model, we further simulate data at daily frequency. We fix the sample size at T , and simulate the data based on (3) in Section 2. For each simulation, we carry out the following steps:

- Step 1: Generate a random sample of $F_1, F_1, \dots, F_T$ with size T from the r -dimensional normal distribution $N(\mu_F, \text{cov}_F)$, where $\mu_F = b$ and cov_F is a diagonal matrix. The diagonal elements of cov_F are $\kappa \odot \beta$.
- Step 2: Take $d \times r$ factor loading matrix $\mathbf{B}$ as β .
- Step 3: Generate d standard deviations $\psi_1, \dots, \psi_d$ of the errors as a random sample of size d from a gamma distribution $G(\alpha_0, \beta_0)$, where $\alpha_0 = 3.3586$ and $\beta_0 = 0.1876$.
- Step 4: After getting d standard deviations $\psi_1, \dots, \psi_d$ of the errors, we generate a random sample of $Z = (Z_1, Z_2, \dots, Z_d)^\top$ with size T from the d -variate normal distribution $N(0, \Psi)$.
- Step 5: Then from equation (3), we calculate the return $y_\tau, j = 1, 2, \dots, r, \tau = 1, 2, \dots, T$.

We choose: $r = 3, \kappa = (3, 4, 5), \theta = (0.09, 0.04, 0.06), \eta = (0.3, 0.4, 0.3), \rho = (-0.6, -0.4, -0.25), b = (0.05, 0.03, 0.02), \beta_1 \sim \mathcal{U}[0.25, 2.25], \beta_2, \beta_3 \sim \mathcal{U}[-0.5, 0.5]$ and $\sigma_{MN} = 0.05$. The variances on the diagonal of Γ are uniformly generated from $[0.2, 0.5]$, with a constant within-block correlation of 0.25. In total, there are 10 blocks of equal size on the diagonal of the residual covariance matrix. We set various d and T to investigate the influence of dimensionality and low-frequency sample size. We estimate the covariance matrix based on data sampled at various frequencies using the previous-tick approach from a 21-day interval, with each day having 6.5 trading hours. We also compare with the case without microstructure noise and asynchronous observations. Steps 2 and 3 guarantee the linkage between the generated high-frequency and low-frequency data.

4.2. Simulation results

In tables 1–6, we compare the performance of our estimators in (13) with the estimators proposed by Fan *et al.* (2016), that is, HFFBC, which can be denoted as

$$\widehat{\Sigma}^{\text{HFFBC}} = \widehat{\beta} \widehat{\beta}^\top + \widehat{\Gamma}^S, \quad (18)$$

where $\hat{\beta}$ is the factor loading matrix in the high-frequency regression, which can be calculated as

$$\hat{\beta} = \hat{\Pi}^{12}(\hat{\Pi}^{22})^{-1}. \quad (19)$$

We present the errors with data contaminated by microstructure effects and data free from such effects, that is, $\|\hat{\Sigma} - \Sigma\|_{\text{MAX}}$, where $\hat{\Sigma}$ denotes either MFACE or HFFBC. Each column represents different high-frequency sampling frequencies, decreasing from left to right. Each row represents different sample lengths of low-frequency data ($T = 500, 1000, 2000$), increasing from top to bottom. From bottom left to top right, Δ_n/T shows an increasing trend.

For covariance matrix estimation, we find that when moderate sampling frequencies are adopted ($\Delta_n = 1$ m to $\Delta_n = 30$ m), the estimation error of the precision matrix estimation is smaller when the low-frequency sample size is moderate, consistent with (15). However, when the dimension is low ($d = 30$), this conclusion is not apparent. Overall, no clear relationship between Δ_n and T is observed. Except for individual cases, such as $\Delta_n = 30$ s, $d = 60$, the estimation accuracy of the MFACE covariance matrix is superior to that of the HFFBC covariance matrix estimation. This result is particularly evident at $\Delta_n = 5$ m and $\Delta_n = 15$ m, which are suggested as the optimal sampling frequencies in the related research. Also, comparing the results of tables 1, 3, and 5 ($d = 30, 60, 120$), we find that the estimation error of the covariance matrix increases with the increase in asset dimension, consistent with (15).

For precision matrix estimation, with the increase in sampling frequency (Δ_n decreases) and the increase in low-frequency sample size, the estimation error of the precision matrix estimation of MFACE decreases, consistent with (17). The estimation accuracy of the MFACE precision matrix is superior to that of the HFFBC precision matrix. This result is particularly evident at $\Delta_n = 5$ m, $\Delta_n = 15$ m, and $\Delta_n = 1$ h. Comparing the results of tables 2, 4, and 6 ($d = 30, 60, 120$), we find that the estimation error of the precision matrix increases with the increase in asset dimension, consistent with (17). When the dimension is low, the estimation error of the precision matrix does not seem to be affected by market microstructure effects.

Overall, the errors of the estimators MFACE in both cases of considering and not considering market microstructure increase with the dimension, consistent with Theorem 1 in this paper and Theorem 1 of Fan *et al.* (2016). Given the frequency and sample length, MFACE is generally superior to HFFBC in both the estimation error of the covariance matrix estimation in the max norm and the precision matrix estimation in the operator norm, indicating that the estimation performance of MFACE is overall better than HFFBC. This conclusion holds true for both cases of considering and not considering market microstructure. When the data generation process considers market microstructure, the error is larger than when it does not, as expected. In both cases, the error increases with the increase in sampling frequency, which may be due to information loss caused by sub-sampling.

The errors of our estimator are smaller in both cases, respectively. The errors of both estimators increase with the dimensionality, in accord with Theorem 1 in this paper and

Theorem 1 in Fan *et al.* (2016). The errors of both cases also increase against the sampling frequency, which may result from the loss of information due to subsampling. We notice that the errors do not seem to be affected by the market microstructure effect when the sampling frequency is relatively high, which indicates that the dimensionality dominates compared with the market microstructure effect.

4.3. Results under different noise setup

To check whether the simulation results are sensitive to different setups, we conduct two groups of experiments with different microstructure noise processes. We first simulate the noise with normal random noise with mean zero and variance $\sigma_{MN}^2 = 0.01^2$. We also simulate the noise in log prices for each asset as an MA(1) process, i.e. $\epsilon_{i,t_j}^x = \xi_{i,t_j} - 0.5\xi_{i,t_{j-1}}$, where ξ_{i,t_j} is an i.i.d. normal noise with mean 0 and variance 0.001^2 . The setting accounts for the microstructure noise in terms of bid-ask spread, justified by Roll (1984). Dai *et al.* (2019) adopted a similar setup in the simulation part. In each group of experiments, we fix $d = 60$ and $T = 1000$, a moderate combination of asset dimension and low-frequency data length.

The results in tables 7 and 8 verify the robustness of different noise setup. In both Case 1 and 2, the estimation errors of HFFBC are larger than those of MFACE given the sampling frequency. This agrees with the main results in section 4.2. Compared with tables 3 and 4, we further find that less volatile noises result in smaller errors and dependence structure in the noise have little impact on the simulation results.

5. Empirical analysis

We apply the MFACE approach to SP500 (Standard&Poor 500 Index) components to assess its empirical performance. Here we focus on the performance in terms of out-of-sample forecasts. In empirical applications, the ‘true’ covariances are unobservable. For this reason we focus on forecasts instead of covariance measurements. We do this in two ways. First, we evaluate the forecasting performance of the MFACE and three other estimators for the volatility of vast dimensional equally-weighted portfolios. Second, we compare the out-of-sample performance by constructing minimum variance portfolios.

5.1. Data

Pioneering work on the factor model includes Sharpe (1976), Lintner (1965), Ross (1976), Chen *et al.* (1986). Chamberlain and Rothschild (1983) extended the classical factor model in the arbitrage pricing theory (APT) to the approximate factor model, in which the covariance matrix of the residual can be off-diagonal. Among the empirical models in an attempt to explain security yields, the three-factor model proposed by Fama and French (1993) is one of the most common models. The factors in their research are based on the portfolio which is constructed by sorting the characteristics of the company. In this section, we use Fama-French three factors as observe

Table 1. Simulation results of covariance estimates when $d=30$.

	$\Delta_n = 30 \text{ s}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 5 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 15 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 30 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ h}$ ($\times 10^{-2}$)
Case 1: with microstructure						
MFACE						
$T = 500$	2.9558	2.9925	3.3127	3.8946	4.6041	5.7458
$T = 1000$	2.9689	2.9977	3.3199	3.9017	5.7363	5.7362
$T = 2000$	2.9690	3.0004	3.3287	3.9196	4.6042	5.7580
HFFBC	2.9792	3.0052	3.3551	4.2142	5.2843	6.8993
Case 2: without microstructure						
MFACE						
$T = 500$	2.9531	2.9903	3.3125	3.8963	4.6047	5.7452
$T = 1000$	2.9651	2.9954	3.3206	3.9035	5.7358	5.7358
$T = 2000$	2.9649	2.9979	3.3288	3.9215	4.6048	5.7573
HFFBC	2.9745	3.0025	3.3542	4.2159	5.2846	6.8985

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

Table 2. Simulation results of precision matrix estimates when $d = 30$.

	$\Delta_n = 30 \text{ s}$	$\Delta_n = 1 \text{ m}$	$\Delta_n = 5 \text{ m}$	$\Delta_n = 15 \text{ m}$	$\Delta_n = 30 \text{ m}$	$\Delta_n = 1 \text{ h}$
Case 1: with microstructure						
MFACE						
$T = 500$	3.7619	3.8062	4.0330	4.6375	5.4275	7.0271
$T = 1000$	3.7586	3.8001	4.0235	4.6191	7.0169	7.0169
$T = 2000$	3.7547	3.7999	4.0337	4.6218	5.4278	7.0340
HFFBC	3.7579	3.8031	4.0622	4.7212	5.6059	7.2783
Case 2: without microstructure						
MFACE						
$T = 500$	3.7790	3.8164	4.0358	4.6364	5.4274	7.0295
$T = 1000$	3.7705	3.8099	4.0266	4.6179	7.0192	7.0192
$T = 2000$	3.7688	3.8099	4.0369	4.7190	5.4273	7.0366
HFFBC	3.7701	3.8133	4.0660	4.7189	5.6059	7.2821

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

Table 3. Simulation results of covariance estimates when $d = 60$.

	$\Delta_n = 30 \text{ s}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 5 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 15 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 30 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ h}$ ($\times 10^{-2}$)
Case 1: with microstructure						
MFACE						
$T = 500$	5.4859	5.4974	5.7809	6.4326	7.1972	8.3367
$T = 1000$	5.4690	5.4820	5.7689	6.4169	7.1618	8.2801
$T = 2000$	5.4690	5.4872	5.7626	6.4211	7.1698	8.2890
HFFBC	5.4615	5.4910	5.9196	6.6193	7.6398	9.1609
Case 2: without microstructure						
MFACE						
$T = 500$	5.4710	5.4895	5.7793	6.4328	7.1968	8.3380
$T = 1000$	5.4546	5.4735	5.7673	6.4171	7.1609	8.2814
$T = 2000$	5.4546	5.4790	5.7614	6.4217	7.1706	8.2903
HFFBC	5.4466	5.4832	5.9189	6.6195	7.6383	9.1620

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

factors. Momentum factor will be considered in section 5.6 as the robustness check.

Our datasets comprise the constituents of the S&P500. We only use those stocks that were included in the index during the complete sample period, which runs from Jan 2, 2004 until Dec 30, 2016. This leaves 248 stocks with complete data. To

balance the competing objectives of maximizing data utilization while addressing the challenges of market microstructure noise and asynchronous returns, we adopt 5-minute return intervals for high-frequency data, as recommended by the simulations presented above. For low-frequency data, we choose daily frequency.

Table 4. Simulation results of precision matrix estimates when $d = 60$.

	$\Delta_n = 30 \text{ s}$	$\Delta_n = 1 \text{ m}$	$\Delta_n = 5 \text{ m}$	$\Delta_n = 15 \text{ m}$	$\Delta_n = 30 \text{ m}$	$\Delta_n = 1 \text{ h}$
Case 1: with microstructure						
MFACE						
$T = 500$	6.5891	6.6118	6.7738	7.4516	8.7313	11.5689
$T = 1000$	6.5839	6.6099	6.7686	7.4416	8.7262	11.5703
$T = 2000$	6.5839	6.6085	6.7667	7.4411	8.7273	11.5645
HFFBC	6.5883	6.6153	6.7980	7.5147	8.8501	11.8018
Case 2: without microstructure						
MFACE						
$T = 500$	6.5855	6.6128	6.7757	7.4492	8.7302	11.5690
$T = 1000$	6.5798	6.6124	6.7705	7.4391	8.7251	11.5698
$T = 2000$	6.5798	6.6107	6.7686	7.4436	8.7262	11.5644
HFFBC	6.5858	6.6176	6.7992	7.5120	8.8486	11.7999

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

Table 5. Simulation results of covariance estimates when $d = 120$.

	$\Delta_n = 30 \text{ s}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 5 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 15 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 30 \text{ m}$ ($\times 10^{-2}$)	$\Delta_n = 1 \text{ h}$ ($\times 10^{-2}$)
Case 1: with microstructure						
MFACE						
$T = 500$	11.0297	11.0509	11.3848	11.9698	12.5519	14.8941
$T = 1000$	11.0190	11.0478	11.3657	11.9395	12.5232	14.1204
$T = 2000$	11.0200	11.0432	11.3742	11.9437	12.5261	14.0870
HFFBC	11.0235	11.0500	11.5063	12.1293	13.0115	14.1266
Case 2: without microstructure						
MFACE						
$T = 500$	11.0131	11.0428	11.3849	11.9706	12.5520	14.8938
$T = 1000$	11.0023	11.0397	11.3655	11.9405	12.5232	14.1199
$T = 2000$	11.0033	11.0353	11.3741	11.9448	12.5257	14.0866
HFFBC	11.0062	11.0418	11.5064	12.1300	13.0109	14.1271

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

Table 6. Simulation results of precision matrix estimates when $d = 120$.

	$\Delta_n = 30 \text{ s}$	$\Delta_n = 1 \text{ m}$	$\Delta_n = 5 \text{ m}$	$\Delta_n = 15 \text{ m}$	$\Delta_n = 30 \text{ m}$	$\Delta_n = 1 \text{ h}$
Case 1: with microstructure						
MFACE						
$T = 500$	10.0997	10.1061	10.2189	10.9534	13.0690	18.7670
$T = 1000$	10.0953	10.1012	10.2131	10.9459	13.0669	18.7649
$T = 2000$	10.0931	10.0994	10.2116	10.9428	13.0604	18.7592
HFFBC	10.0969	10.1067	10.2284	10.9953	13.1918	18.9174
Case 2: without microstructure						
MFACE						
$T = 500$	10.0900	10.1010	10.2197	10.9547	13.0734	18.7596
$T = 1000$	10.0841	10.0959	10.2145	10.9468	13.0713	18.7577
$T = 2000$	10.0815	10.0942	10.2128	10.9443	13.0648	18.7522
HFFBC	10.0847	10.1016	10.2293	10.9963	13.1962	18.9089

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

We make use of observable factors constructed from high-frequency returns, including the market portfolio, the small-minus-big market capitalization (SMB) portfolio, and high-minus-low price-earnings ratio(HML) portfolio in the Fama-French 3 factor model, as well as the daily-rebalanced momentum portfolio formed by sorting stock returns between the past 250 days and 21 days. These factors are constructed

by adapting the Fama-French procedure to a high frequency setting (see Aït-Sahalia et al. 2020).

In terms of the information set S mentioned in section 2, Fan et al. (2016), Ait-Sahalia and Xiu (2017) both took advantage of the GICS (Global Industry Classification Standard) information. We also use GICS in the empirical analysis. Figure 2 illustrates the number of stocks in each category.

Table 7. Simulation results under of covariance estimates when $d = 60$ and $T = 1000$.

	$\Delta_n = 30$ s	$\Delta_n = 1$ m	$\Delta_n = 5$ m	$\Delta_n = 15$ m	$\Delta_n = 30$ m	$\Delta_n = 1$ h
Case 1: 0.01						
MFACE	5.4534	5.4741	5.7674	6.4170	7.1610	8.2811
HFFBC	5.4473	5.4836	5.9187	6.6193	7.6386	9.1618
Case 2: MA noise						
MFACE	5.5464	5.4760	5.7677	6.4169	7.1613	8.2807
HFFBC	5.4505	5.4853	5.9188	6.6192	7.6390	9.1615

Note: We conduct each simulation for 100 times and report the mean. 1m, 5m, 15m, 30m and 65m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

Table 8. Simulation results of precision matrix estimates when $d = 60$ and $T = 1000$.

	$\Delta_n = 30$ s	$\Delta_n = 1$ m	$\Delta_n = 5$ m	$\Delta_n = 15$ m	$\Delta_n = 30$ m	$\Delta_n = 1$ h
Case 1: 0.01						
MFACE	6.5832	6.6125	6.7703	7.4399	8.7256	11.5701
HFFBC	6.5860	6.6177	6.7992	7.5128	8.8491	11.8005
Case 2: MA noise						
MFACE	6.5835	6.6120	6.7698	7.4408	8.7261	11.5703
HFFBC	6.5861	6.6172	6.7990	7.5138	8.8497	11.8001

Note: We conduct each simulation for 100 times and report the mean. 1 m, 5 m, 15 m, 30 m and 65 m denote that the sampling frequencies are 1 minute, 5 minutes, 15 minutes, 30 minutes and 65 minutes respectively.

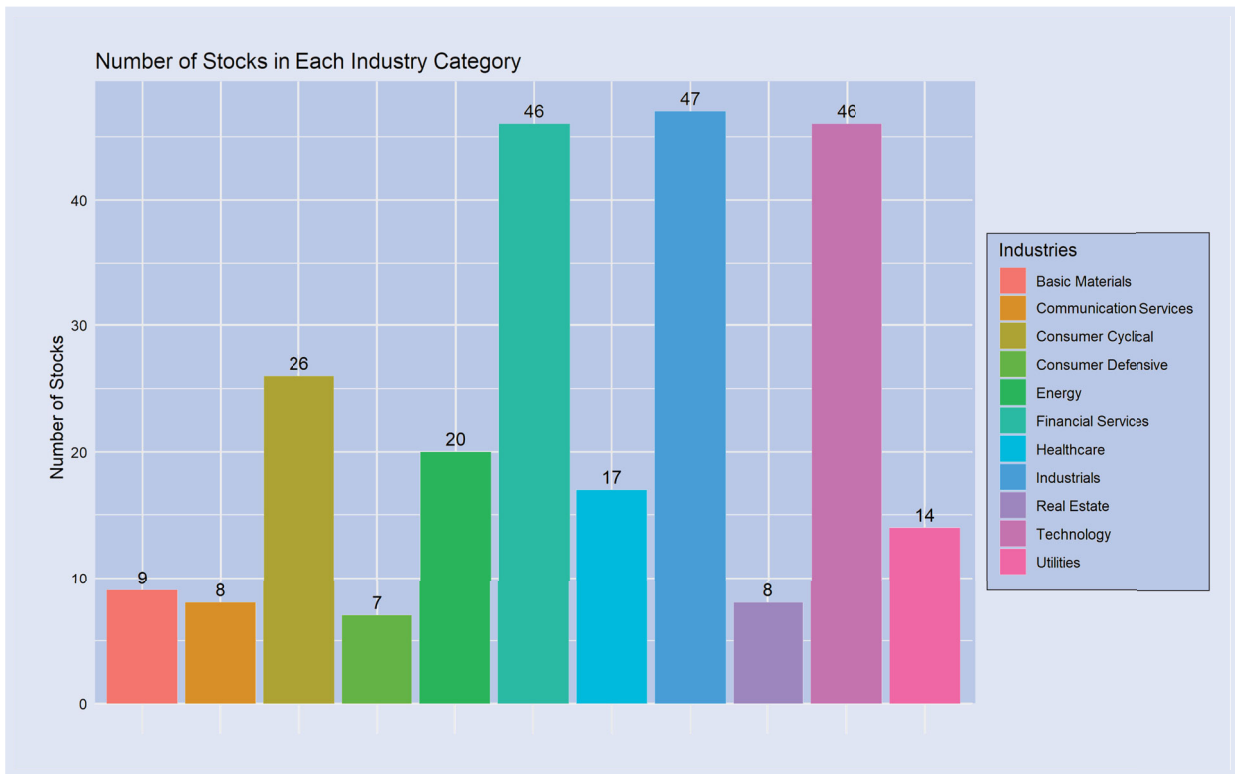


Figure 2. Number of stocks in each industry category.

5.2. Covariance matrix estimators and forecasts

We compare the performance of our estimator (MFACE) in the portfolio analysis with three other estimators: realized covariance (RC), low-frequency factor-based covariance estimator (LFFBC), realized mixed-frequency factor models (MFFM) proposed by Bannouh *et al.* (2012), and the estimator (HFFBC) proposed by Fan *et al.* (2016). Factor structure

is imposed in all the above estimators except the realized covariance estimator.

Realized covariance based on intraday returns can be computed as:

$$\widehat{\Sigma}^{\text{RC}} = \frac{1}{t} \sum_{j=1}^{\lfloor t/\Delta_n \rfloor} (\Delta_j^n Y)(\Delta_j^n Y)^\top. \quad (20)$$

Low-frequency factor-based covariance estimator can be denoted as:

$$\widehat{\Sigma}^{\text{LFFBC}} = \widehat{\mathbf{B}}\widehat{\text{cov}}(\mathbf{f})\widehat{\mathbf{B}}^\top + \widehat{\text{cov}}(\mathbf{u}), \quad (21)$$

where $\widehat{\text{cov}}(\mathbf{f})$ and $\widehat{\text{cov}}(\mathbf{u})$ are the sample covariance of daily factors and residuals $\mathbf{f}$ and $\mathbf{u}$. For the MFFM approach, we first run regression on daily data to obtain the factor loadings $\widehat{\mathbf{B}}$. The intraday residuals needed to compute idiosyncratic variances are obtained as

$$\hat{\epsilon} = \Delta Y - \widehat{\mathbf{B}}\Delta F, \quad (22)$$

where ΔY and ΔF are corresponding high-frequency returns of assets and factors. Finally the MFFM covariance matrix estimator is then computed as

$$\widehat{\Sigma}^{\text{MFFM}} = \widehat{\mathbf{B}}\widehat{\mathbf{E}}\widehat{\mathbf{B}}^\top + \text{diag}(\hat{\epsilon}\hat{\epsilon}^\top). \quad (23)$$

And the estimator proposed by Fan *et al.* (2016) is given in equations (19) and (20).

We consider forecasts based on an exponentially weighted moving average (EWMA) scheme, motivated by the work of Foster and Nelson (1996), Andreou and Ghysels (2002). In this framework, the covariance matrix forecast for day τ is given by

$$\Sigma_{\tau|\tau-1} = \alpha \Sigma_{\tau-1|\tau-2} + (1 - \alpha) \hat{\Sigma}_{\tau-1}, \quad (24)$$

where the scalar α is a fixed decay parameter and $\hat{\Sigma}_{\tau}$ is the covariance matrix estimate for day $\tau - 1$ as given by each of the four estimators. We consider several weighting schemes with $\alpha \in \{0.94, 0.75, 0.50, 0.25\}$. The value of 0.94 for α is the optimal decay parameter for daily data documented by RiskMetrics. The use of smaller decay parameters allows us to examine the effects on the forecasting performance when putting more weight on more recent data. Further, using smaller values of α provides more insight into the quality of the covariance estimator itself rather than the smoothed forecast. Less smoothing can be important also from an economic point of view, as it enables the forecasts to adjust more rapidly to important changes in (co)variance dynamics, which for example occur at turning points between periods of high and low volatility. It is worth mentioning that when $\alpha = 0$, EWMA scheme is equal to random walk forecasting approach, which is adopted by Fan *et al.* (2016) and Dai *et al.* (2019).

We use the data from January 2, 2004 to December 30, 2016 for the covariance dynamics in (24). The lengths of the moving windows for daily and intraday data are $l_0 = 504$ and $TT = 22$ trading days below the forecast date, approximately the trading days in two years and a month, respectively. The out-of-sample period runs from January 3, 2006 until December 30, 2016, the length of which is denoted as l_1 .

5.3. Equally-weighted portfolios

In our first forecasting exercise, we consider equally-weighted portfolios for the SP500 stock universes. As noted before, we only use the constituents that were included in a single

Table 9. SP500 equally-weighted portfolio volatility(daily squared return), Mincer–Zarnowitz regressions.

	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 0.94$
RC	0.2947	0.2871	0.2800
LFFBC	0.0271	0.0268	0.0264
MFFM	0.3169	0.3097	0.3030
HFFBC	0.2949	0.2873	0.2802
MFACE	0.3089	0.3027	0.2971

Table 10. SP500 equally-weighted portfolio volatility (realized volatility), Mincer–Zarnowitz regressions.

	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 0.94$
RC	0.4933	0.4809	0.4694
LFFBC	0.0494	0.0483	0.0474
MFFM	0.5134	0.5036	0.4944
HFFBC	0.4935	0.4811	0.4696
MFACE	0.5064	0.4988	0.4915

index during the complete sample period. For each universe the daily equally-weighted portfolio return is computed as $y_\tau^p = e^\top y_\tau$, where y_τ is a vector of close-to-close returns on the individual stocks and e is the equal-weight vector with entries $1/d$. We obtain one-day ahead forecasts of the volatility of these equally-weighted portfolios as $\hat{\sigma}_{\tau|\tau-1}^p = e^\top \Sigma_{\tau|\tau-1} e$, using the covariance matrix forecasts based on the four types of estimators.

We run Mincer–Zarnowitz (MZ) regressions, in which the portfolio volatility proxy $(\hat{\sigma}_{\tau|\tau-1}^p)^2$ is regressed on a constant and one of the volatility forecasts, that is,

$$(\hat{\sigma}_{\tau|\tau-1}^p)^2 = \gamma + \delta(\hat{\sigma}_{\tau|\tau-1}^p)^2 + \epsilon_\tau. \quad (25)$$

We first use the squared daily return $(y_\tau^p)^2$ as the volatility proxy. Although this proxy is known to be noisy, at least it is unbiased. Then we adopt Realized Volatility (RV) as the volatility proxy.

Tables 9 and 10 presents the R^2 for the Mincer–Zarnowitz regressions. We find that the two mixed-frequency estimators, MFFM and MFACE, performs best among the five. Considering that MFFM behaves poorly in our empirical analysis due to the singularity of the low-rank part $(\widehat{\mathbf{B}}\widehat{\mathbf{E}}\widehat{\mathbf{B}}^\top)$, MFACE does a good job in forecasting. Also, we find that the two mixed-frequency estimators are robust under different decay parameters, which is evidenced by the slight difference in the regression R^2 s. Furthermore, R^2 's show significant improvements when using RV instead of the squared daily return.

To compare the forecast abilities of different models, we adopt the Model Confidence Set (MCS, Hansen *et al.* 2011). The MCS consists of a sequence of statistic tests which permits to construct a set of ‘superior’ models, the ‘Superior Set Models’ (SSM), where the null hypothesis of equal predictive ability (EPA) is not rejected at certain confidence level α_0 . The EPA statistic tests is evaluated for an arbitrary loss function, which essentially means that it is possible to test models on various aspects depending on the chosen loss function.

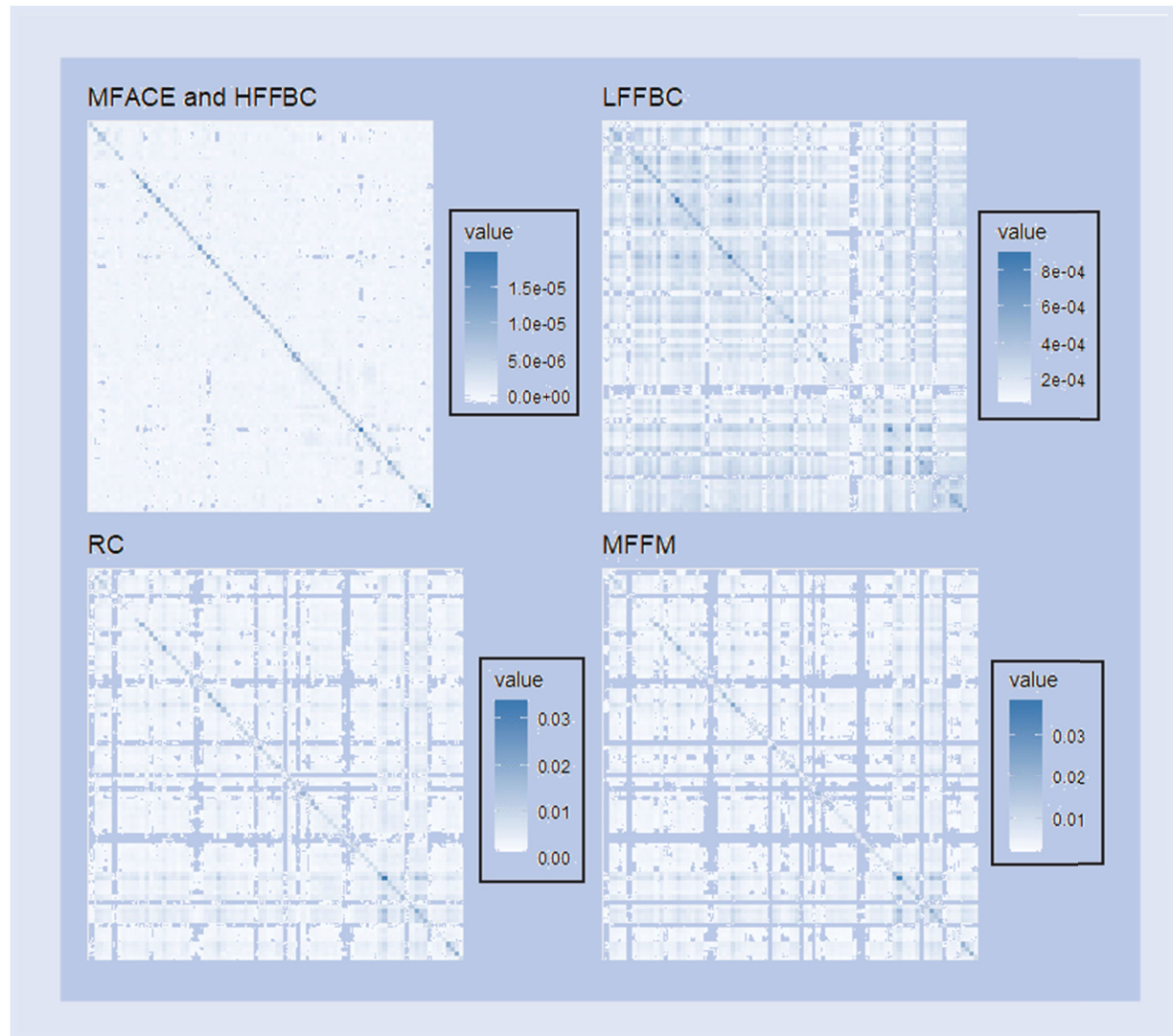


Figure 3. The realized sparsity of residual covariance matrix. *Notes:* The figure shows the heatmap of the average idiosyncratic realized covariances from 2016-11-30 to 2016-12-30. Darker shades indicate higher covariances in absolute value.

Patton (2011) shows that among a collection of loss functions employed in the literature on volatility forecast evaluation, the Mean Squared Error (MSE) and Quasi-Likelihood (QLIKE) are 'robust' and can lead to correct rankings of volatility forecasts under certain assumptions. We choose the MSE loss function, which is defined as $((\hat{\sigma}_{\tau|\tau}^p)^2 - (\hat{\sigma}_{\tau|\tau-1}^p)^2)^2$. We set $\alpha = 0$. When RV is adopted as volatility proxy, MFACE is the only estimator in the SSM, indicating its superior forecast ability. RC predicts squared daily returns more precisely compared to the other four estimators. For $\alpha = 0.5$ and $\alpha = 0.94$, the conclusion also holds.

5.4. Realized residual sparsity

Figure 3 provides the covariance structures of the idiosyncratic component under different strategies. It should be noted that we do not apply any blocking or thresholding technique in this part. MFACE and HFFBC share the same idiosyncratic component. For the realized covariance (RC), the covariance matrix the idiosyncratic component is the realized covariance itself. Compared with Fan *et al.* (2016) and Ait-Sahalia and Xiu (2017), we do not threshold on the residual covariance

matrix, so as to avoid involving additional tuning parameters. In figure 3, we can see the residual sparsity patterns.

The results in figure 3 show several important features. (i) The common components of MFACE and HFFBC estimators have fully extracted the correlation among the assets, and the elements outside the industry blocks, even most off-diagonal elements inside the blocks are close to 0, which demonstrates that the Assumption 2 in this paper is supported by empirical data. (ii) The realized covariance matrix of the idiosyncratic component estimated by MFACE and HFFBC reveals correlation inside the industry, especially in the Energy sector (upper left corner) and the Financial sector (on the lower right next to the Energy sector), which confirms the necessity of imposing approximate factor structure to the asset covariance matrix. In contrast, MFFM only retains the diagonal elements of the residual covariance matrix, which results in the misspecification bias, that is, the loss of information in the asset covariance matrix which has not been extracted by the strict factor structure. (iii) Compared with the estimators taking advantage of high-frequency data, LFFBC estimator only utilizes low-frequency information. In this respect, the common factors do not effectively extract the correlation

structure of the asset covariance matrix due to the lack of information.

5.5. Minimum variance portfolios

Fan *et al.* (2008) proved that the precision matrix with factor structure is better than the sample precision matrix in the convergence rate, which may have a significant impact on the minimum variance portfolio selection and other occasions involving the precision matrix estimation. One of the main concerns in this paper is whether imposing approximate factor structure to the covariance matrix based on mixed-frequency data will improve the performance of minimum variance portfolio. Given the one-day ahead EWMA forecasts of the covariance matrices, we construct the standard fully-invested minimum variance portfolios (MVP):

$$\omega_\tau = \frac{\Sigma_{\tau|\tau-1}^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma_{\tau|\tau-1}^{-1} \mathbf{1}}, \quad (26)$$

where $\mathbf{1}$ is a d -dimensional vector of ones. The portfolio return can be calculated by $y_\tau^{mvp} = \omega_\tau^\top y_\tau$.

We first consider the minimum risk of the portfolio. Combining (26), the annualized minimum risk of the portfolio can be expressed as:

$$AR = \frac{250}{l_1} \sqrt{\omega_\tau^\top \Sigma_{\tau|\tau-1} \omega_\tau} = \frac{250}{l_1} \frac{1}{\sqrt{\mathbf{1}^\top \Sigma_{\tau|\tau-1}^{-1} \mathbf{1}}}. \quad (27)$$

We also keep track of the daily turnover and leverage in the portfolio weights ω_τ which are directly associated with the transaction cost of an investor who wishes to re-balance his or her portfolio daily. We compute turnover by summing the absolute daily weight changes over the stocks,

$$TO_\tau = |\omega_\tau - \omega_{\tau-1}|^\top \mathbf{1}. \quad (28)$$

We compute average (daily) gross leverage as

$$GL = \frac{1}{l_0} |\omega_\tau|^\top \mathbf{1} \quad (29)$$

We compute average (daily) proportion of leverage as

$$PL = \frac{1}{l_0 \times N} \sum_{k=1}^{240} \sum_{i=1}^d 1_{\{\omega_{i,\tau} < 0\}} \quad (30)$$

where $1_{\{\cdot\}}$ denotes the indicator function. Moreover, we consider two tail risk measures, Value at Risk (VaR) and Expected Shortfall (ES) at confidence level 0.05. VaR represents the potential loss in value of a portfolio over a specific time frame, with a certain level of confidence. It is defined as follows:

$$VaR_{0.95} = -\inf\{z \in \mathbb{R} : P(y_\tau^{mvp} \leq z) \geq 0.05\}, \quad (31)$$

where VaR_α is the Value at Risk at confidence level 0.05, R_p is the return on the portfolio, $P(R_p \leq z)$ is the probability that the portfolio return is less than or equal to z . ES, also known as Conditional Value at Risk (CVaR), is a risk measure that

provides the expected loss in the tail of the loss distribution beyond the VaR. ES is defined as follows:

$$ES_{0.95} = -\frac{1}{0.95} \int_0^{0.95} P(y_\tau^{mvp} \leq z) dz \quad (32)$$

where $ES_{0.95}$ is the Expected Shortfall at confidence level 0.05, R_p is the return on the portfolio, $P(R_p \leq z)$ is the cumulative distribution function of the portfolio return. The behavior of some portfolio performance measures are evaluated in Jobson and Korkie (1981).

As discussed by Bannouh *et al.* (2012), we expect that a covariance estimator that is well-conditioned and numerically stable will result in smaller daily portfolio turnover. The daily turnover will also be related to the decay parameter α in (24) which is used to generate EWMA forecasts. A large decay parameter implies that more weight is assigned to historical estimates whereas a smaller decay parameter corresponds to assigning more weight to the most recent estimate(s). More weight on historical estimates will make an estimator more stable and cause less portfolio turnover but on the other hand, recent shifts will be picked up at a slower pace.

We omit the results of MFFM in view of its singularity in empirical application. In contrast, MFACE behaves well in high-dimensional MVP since the sparsity assumption combining GICS provides more information on the correlation of the idiosyncratic components. The results in table 11 highlight the benefits of incorporating an approximate factor structure into covariance estimators. On the one hand, the two estimators with approximate factor structure (HFFBC and MFACE) attain lower risk levels than that without factor structure (RC) and that imposes no structure on the idiosyncratic components (LFFBC) and display strong robustness to decay parameters. On the other hand, the two estimators (HFFBC and MFACE) based on the approximate factor model not only impose the factor structure, but also restrict the correlation structure of idiosyncratic components inside the sector blocks compared to LFFBC and as such investors need not adjust the portfolio weight frequently, resulting in lower turnovers. Wang *et al.* (2023) illustrates the trade-off between timeliness and low turnover in portfolio construction. Between these two estimators with approximate factor structure, the mixed-frequency data in MFACE can alleviate the effect of noise in high-frequency data with smaller risk, which provides evidence for our discussion in section 2.3. In addition, the turnovers of the minimum variance portfolios display a trend with regard to decay parameters. Turnovers increase dramatically when $\alpha = 0.94$ compared to $\alpha = 0.5$ while the relative relationship among the four estimators remain the same. In practice, the choice of forecasting scheme may have a significant impact on the trading cost in portfolio construction.

Turning now to the results on leverage. MFACE consistently and markedly outperforms all the other models in terms of percentage of leverage (PL). HFFBC consistently outperforms all the other models in terms of gross leverage (GL). Using mixed-frequency data instead of high-frequency or low-frequency data alone in the estimation of the factor models consistently reduces annualized risks, turnover and percentage of leverage, but has no noticeable effect on

Table 11. Annualized risk, average daily turnover and leverage of MVP.

	AR($\times 10^{-4}$)	TO($\times 10^2$)	PL	GL	VaR _{0.95}	ES _{0.95}
A: $\alpha = 0$						
RC	3.2678	23.0636	0.4602	3.8738	-0.0081	-0.0152
LFFBC	2.6369	13.1459	0.4777	6.7644	-0.0099	-0.0121
HFFBC	0.3400	1.4524	0.4613	0.4109	-0.0010	-0.0010
MFACE	0.1249	2.5479	0.1286	1.0743	-0.0143	-0.0249
B: $\alpha = 0.5$						
RC	3.2823	17.0405	0.4601	3.8515	-0.0076	-0.0076
LFFBC	2.6390	10.4095	0.4780	6.7564	-0.0099	-0.0122
HFFBC	0.3406	1.0942	0.4616	0.4097	-0.0007	-0.0007
MFACE	0.1249	1.9130	0.1279	1.0732	-0.0143	-0.0250
B: $\alpha = 0.94$						
RC	3.2715	21.6130	0.4602	3.8680	-0.0069	-0.0069
LFFBC	2.6373	12.2047	0.4779	6.7608	-0.0100	-0.0121
HFFBC	0.3402	1.3727	0.4613	0.4106	-0.0003	-0.0003
MFACE	0.1249	2.4069	0.1284	1.0740	-0.0143	-0.0250

Table 12. Robustness of time span.

	AR ($\times 10^{-4}$)	TO($\times 10^2$)	PL	GL	VaR _{0.95}	ES _{0.95}
A: $L = 252$						
RC	2.8114	27.5242	0.4605	3.9828	-0.0077	-0.0109
LFFBC	0.7232	440.8133	0.4923	29.6898	-0.0403	-0.0741
HFFBC	0.2958	1.6530	0.4602	0.4089	-0.0010	-0.0010
MFACE	0.1087	2.8976	0.1265	1.0745	-0.0140	-0.0249
B: $l_1 = 755$						
RC	13.6733	6.10053	0.4538	3.5292	-0.0110	-0.0110
LFFBC	7.3565	3.6307	0.4765	6.1884	-0.0131	-0.0186
HFFBC	1.4399	0.4154	0.4524	0.4197	-0.0016	-0.0016
MFACE	0.5427	0.6876	0.1064	1.0553	-0.0197	-0.0349
C: $TT = 5$						
RC	0.9225	379.4871	0.4817	11.7340	-0.0180	-0.0421
LFFBC	2.4786	2.9293	0.4696	3.3965	-0.0102	-0.0114
HFFBC	0.2433	7.8896	0.4507	0.5524	-0.0015	-0.0045
MFACE	0.0980	13,8439	0.2008	1.2166	-0.0144	-0.0253
D: $\Delta_n = 15m$						
RC	1.3906	52.5565	0.4662	5.4967	-0.0101	-0.0150
LFFBC	2.6369	13.1459	0.4777	6.7644	-0.0099	-0.0121
HFFBC	0.3168	2.2297	0.4573	0.4572	-0.0007	-0.0007
MFACE	0.1199	4.0562	0.1630	1.1316	-0.0143	-0.0236

turnover gross leverage. The two estimators (HFFBC and MFACE) with approximate factor structure have better performance in the annualized risks, turnovers and leverage of the minimum variance portfolios. There is a trade-off between the two estimators: MVPs based on MFACE bear higher transaction costs, while attains smaller risks, while the portfolios based on HFFBC have lower transaction costs with higher risks at the same time. ‘Mixed-frequency low rank + sparsity’ does a good job in minimum variance portfolio analysis. Additionally, we observe that HFFBC yields the best performance in terms of tail risk measures, while VaR and ES for the other estimators remain within reasonable levels. The minimization of VaR and ES in portfolio selection is another interesting topic, which lies beyond the scope of this paper. For a review of these models and their robust variants, see Fabozzi *et al.* (2010).

5.6. Robustness analysis

To check the robustness of our strategy, we first narrow the length of the moving windows for daily(low-frequency) data L to 252 days, which is approximately the trading days in a year. Next we adjust the out-of-sample period l_1 to 755 days, from January 3, 2006 to December 30, 2008. We also narrow the length of the moving windows for intraday (high-frequency) data (TT) to 5 days, which is approximately the trading days in a week. Finally, we examine a different intraday frequency. Δ_n is altered from 5m to 15m, another recommended sampling frequency according to the simulation results. We only consider the case where $\alpha = 0$ in (24) in this subsection, that is, random walk scheme. When $\alpha = 0.5, 0.94$, the results are similar to the case where $\alpha = 0$. These results are not listed because of the limitation of space, and they

Table 13. Robustness of cross-section.

	AR ($\times 10^{-4}$)	TO ($\times 10^2$)	PL	GL	VaR _{0.95}	ES _{0.95}
A: $d = 100$						
RC	2.8583	6,132.3802	0.4464	3.3952	− 0.0098	− 0.0236
LFFBC	4.0392	375.8573	0.4487	3.2563	− 0.0102	− 0.0252
HFFBC	0.3446	442.0645	0.4317	0.3832	− 0.0015	− 0.0044
MFACE	0.1339	757.2339	0.0880	1.0422	− 0.0139	− 0.0239
B: FF+MOM						
LFFBC	2.4767	1,498.7626	0.4804	7.3119	− 0.0102	− 0.0114
HFFBC	0.3398	151.8601	0.4619	0.3898	− 0.0007	− 0.0007
MFACE	0.1187	265.9444	0.1265	1.0730	− 0.0145	− 0.0262

Note: There is no RC estimator in Panel B (FF+MOM) because the estimator do not impose any factor structure on the covariance.

are available from the authors on request. We add Momentum factors (MOM) to see whether the number of factors could have a large impact on the estimation. In addition, we randomly select 100 stocks among the 269 stocks used in section 5.5 and reconstruct the minimum variance portfolio. We denote the two cases as ‘FF+MOM’ and ‘ $d = 100$ ’ respectively.

The results are reported in tables 12 and 13. MFACE and HFFBC continue to outperform other methods on AR, TO and GL across nearly all cases. Except for HFFBC, the other three estimators show no considerable superiority or inferiority on ES and VaR. In case ‘FF+MOM’, an interesting question is whether the adoption of the selected factors helps. Among the research on testing the number of factors, Connor and Korajczyk (1993) proposed one in an approximate factor model of asset returns. We adapt their procedures to our high-frequency setting. The t-test statistics is 0.392 and the corresponding p-value is 0.347. The null hypothesis of 3 factors is not rejected given the 95% confidence level. We also find that MFACE and HFFBC perform better when the portfolio size decreases. This rather interesting result could be attributed to stock universe. When we limit the SP500 universe to these 100 stocks, less choice for assets directly lead to fewer transactions and less rebalancing. In general, MFACE has the lowest AR and PL for all the MVPs and MFACE has the lowest TO and GL for all the MVPs. This result indicates that high-frequency data combining factor structure are useful in portfolio choice, especially for controlling the risk.

6. Conclusion

In recent years, high-frequency data has been gradually applied to the estimation and prediction of high-dimensional asset return covariance matrix, and has become a hotspot of financial econometrics. Considering the effect of market microstructure, this paper proposes the mixed-frequency factor-based covariance estimator (MFACE), which not only combines the information of high-frequency intraday data and low-frequency daily data, but also allows the correlation between the idiosyncratic components, that is, the approximate factor structure. The consistency of the covariance estimator and the precision matrix estimator is proved theoretically and the corresponding convergence rates are

derived. From the simulation results, we further verify that the estimator proposed in this paper performs better than the HFFBC estimator proposed by Fan *et al.* (2016). In the empirical study, MFACE efficiently extracts the correlation structure between assets. The results of MZ regression demonstrate that the estimator proposed in this paper is better than the realized covariance (RC), LFFBC and HFFBC in most cases. In addition, an investigation into the minimum variance portfolio indicates that the portfolio constructed by MFACE achieves the lowest risk with relatively small transaction cost and leverage.

The MFACE model proposed in this paper provides a new perspective for estimating the covariance matrix of financial assets in the presence of microstructure noise. The covariance matrix can be decomposed via imposing the approximate factor structure. Different components can be estimated with data of different frequencies. Factor loadings can be estimated by daily low-frequency data to alleviate the market microstructure effect while high-frequency factors are applied to promote the accuracy of the estimator. Another interesting question is how to incorporate parametric model, such as the Dynamic Conditional Correlation model (Engle 2002) or the class of Hadamard Exponential models (Bauwens and Otranto 2023), into mixed-frequency covariance estimator. We leave it for future work.

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ORCID

Yifu Wang  <http://orcid.org/0000-0002-8958-4158>

References

- Aöt-Sahalia, Y., Kalnina, I. and Xiu, D., High-frequency factor models and regressions. *J. Econom.*, 2020, **216**, 86–105.
- Ait-Sahalia, Y. and Xiu, D., Using principal component analysis to estimate a high dimensional factor model with high-frequency data. *J. Econom.*, 2017, **201**, 384–399.
- Andreou, E. and Ghysels, E., Rolling-sample volatility estimators: Some new theoretical, simulation and empirical results. *J. Bus. Econ. Stat.*, 2002, **20**, 363–376.
- Bandi, F. and Russell, J., Microstructure noise, realized variance, and optimal sampling. *Rev. Econ. Stud.*, 2008, **75**, 339–369.
- Bannouh, K., Martens, M., Oomen, R. and van Dijk, D., Realized mixed-frequency factor models for vast dimensional covariance estimation. *ERIM Report Series Reference No. ERS-2012-017-F&A*, 2012.
- Bauwens, L. and Otranto, E., Modeling realized covariance matrices: A class of Hadamard exponential models. *J. Financ. Econom.*, 2023, **21**, 1376–1401.
- Bickel, P.J. and Levina, E., Covariance regularization by thresholding. *Ann. Stat.*, 2008, **36**, 2577–2604.
- Black, F., Noise. *J. Finance*, 1986, **41**, 529–543.
- Bollerslev, T., Meddahi, N. and Nyawa, S., High-dimensional multivariate realized volatility estimation. *J. Econom.*, 2019, **212**, 116–136.
- Cai, T.T., Hu, J., Li, Y. and Zheng, X., High-dimensional minimum variance portfolio estimation based on high-frequency data. *J. Econom.*, 2020, **214**, 482–494.
- Chamberlain, G. and Rothschild, M., Arbitrage, factor structure, and mean-variance analysis on large asset markets. *Econometrica*, 1983, **51**, 1281–1304.
- Chan, L.K.C., Karceski, J. and Lakonishok, J., On portfolio optimization: Forecasting covariances and choosing the risk model. *Rev. Financ. Stud.*, 1999, **12**, 937–974.
- Chen, N., Roll, R. and Ross, S., Economic forces and the stock market. *J. Bus.*, 1986, **59**, 383–403.
- Chen, D., Mykland, P.A. and Zhang, L., Realized regression with asynchronous and noisy high frequency and high dimensional data. *J. Econom.*, 2024, **239**, 105446.
- Connor, G. and Korajczyk, R.A., A test for the number of factors in an approximate factor model. *J. Finance*, 1993, **48**, 1263–1291.
- Dai, C., Lu, K. and Xiu, D., Knowing factors or factor loadings, or neither? Evaluating estimators of large covariance matrices with noisy and asynchronous data. *J. Econom.*, 2019, **208**, 43–79.
- De Nard, G., Ledoit, O. and Wolf, M., Factor models for portfolio selection in large dimensions: The good, the better and the ugly. *J. Financ. Econom.*, 2021, **19**, 236–257.
- Engle, R., Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *J. Bus. Econ. Stat.*, 2002, **20**, 339–350.
- Epps, T., Comovements in stock prices in the very short run. *J. Am. Stat. Assoc.*, 1979, **74**, 291–298.
- Fabozzi, F.J., Huang, D. and Zhou, G., Robust portfolios: Contributions from operations research and finance. *Ann. Oper. Res.*, 2010, **176**, 191–220.
- Fama, E. and French, K., Common risk factors in the returns on stocks and bonds. *J. Am. Stat. Assoc.*, 1993, **33**, 3–56.
- Fan, J., Fan, Y. and Lv, J., High-dimensional covariance matrix estimation using a factor model. *J. Econom.*, 2008, **147**, 186–197.
- Fan, J., Liao, Y. and Mincheva, M., High-dimensional covariance matrix estimation in approximate factor models. *Ann. Stat.*, 2011, **39**, 3320–3356.
- Fan, J., Li, Y. and Yu, K., Vast volatility matrix estimation using high-frequency data for portfolio selection. *J. Am. Stat. Assoc.*, 2012, **107**, 412–428.
- Fan, J., Furger, A. and Xiu, D., Incorporating global industrial classification standard into portfolio allocation: A simple factor-based large covariance matrix estimator with high-frequency data. *J. Bus. Econ. Stat.*, 2016, **34**, 489–503.
- Foster, D. and Nelson, D., Continuous record asymptotics for rolling sample variance estimators. *Econometrica*, 1996, **64**, 139–174.
- Halbleib, R. and Voev, V., Forecasting covariance matrices: A mixed approach. *J. Financ. Econom.*, 2016, **14**, 383–417.
- Hansen, P.R. and Lunde, A., Realized variance and market microstructure noise. *J. Bus. Econ. Stat.*, 2006, **24**, 127–161.
- Hansen, P.R., Lunde, A. and Nason, J.M., The model confidence set. *Econometrica*, 2011, **79**, 453–497.
- Jagannathan, R. and Ma, T., Risk reduction in large portfolios: Why imposing the wrong constraints helps. *J. Finance*, 2003, **58**, 1651–1683.
- Jobson, J.D. and Korkie, B.M., Performance hypothesis testing with the Sharpe and Treynor measures. *J. Finance*, 1981, **36**, 889–908.
- Kim, D., Liu, Y. and Wang, Y., Large volatility matrix estimation with factor-based diffusion model for high-frequency financial data. *Bernoulli*, 2018, **24**, 3657–3682.
- Kim, D., Oh, M., Song, M. and Wang, Y., Dynamic realized beta models using robust realized integrated beta estimator. *arXiv preprint arXiv:2204.06914*, 2022.
- Lintner, J., Security prices, risk, and maximal gains from diversification. *J. Finance*, 1965, **20**, 587–615.
- Lo, A. and MacKinlay, A., An econometric analysis of nonsynchronous trading. *J. Econom.*, 1990, **45**, 181–211.
- Markowitz, H., Portfolio selection. *J. Finance*, 1952, **7**, 77–91.
- Merton, R.C., On estimating the expected return on the market: An exploratory investigation. *J. Financ. Econ.*, 1980, **8**, 323–361.
- Patton, A.J., Volatility forecast comparison using imperfect volatility proxies. *J. Econom.*, 2011, **160**, 246–256.
- Pelger, M., Large-dimensional factor modeling based on high-frequency observations. *J. Econom.*, 2019, **208**, 23–42.
- Peng, S., Guo, S. and Long, Y., Large dimensional portfolio allocation based on a mixed frequency dynamic factor model. *Econom. Rev.*, 2022, **41**, 539–563.
- Petukhina, A., Klochov, Y., Härdle, W.K. and Zhivotovskiy, N., Robustifying markowitz. *J. Econom.*, 2024, **239**, 105387.
- Roll, R., A simple implicit measure of the effective bid-ask spread in an efficient market. *J. Finance*, 1984, **39**, 1127–1139.
- Ross, S., Options and efficiency. *Q. J. Econ.*, 1976, **90**, 75–89.
- Sharpe, W., A simplified model for portfolio analysis. *Manage. Sci.*, 1976, **9**, 277–293.
- Tao, M., Wang, Y., Yao, Q. and Zou, J., Large volatility matrix inference via combining low-frequency and high-frequency approaches. *J. Am. Stat. Assoc.*, 2011, **106**, 1025–1040.
- Tao, M., Wang, Y. and Chen, X., Fast convergence rates in estimating large volatility matrices using high-frequency financial data. *Econ. Theory*, 2013a, **29**, 838–856.
- Tao, M., Wang, Y. and Zhou, H.H., Optimal sparse volatility matrix estimation for high-dimensional Itô processes with measurement errors. *Ann. Stat.*, 2013b, **41**, 1816–1864.
- Wang, Y. and Zou, J., Vast volatility matrix estimation for high-frequency financial data. *Ann. Stat.*, 2010, **38**, 943–978.
- Wang, Y., Lu, W. and Boudt, K., Dynamic core-satellite investing using higher order moments: An explicit solution. *Quant. Finance*, 2023, **23**, 1815–1831.
- Zhang, L., Mykland, P. and Ait-Sahalia, Y., A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *J. Am. Stat. Assoc.*, 2005, **100**, 1394–1411.

Appendices

Appendix 1. Proof of Theorem 1

$$\begin{aligned}
& \|\hat{\Sigma}^{\text{MFACE}} - \Sigma\|_{\text{MAX}} \\
& \leq \|2(\hat{\mathbf{B}} - \mathbf{B})\mathbf{E}\mathbf{B}^{\top}\|_{\text{MAX}} + \|\mathbf{B}(\hat{\mathbf{E}} - \mathbf{E})\mathbf{B}^{\top}\|_{\text{MAX}} \\
& \quad + \|(\hat{\mathbf{B}} - \mathbf{B})\mathbf{E}(\hat{\mathbf{B}} - \mathbf{B})^{\top}\|_{\text{MAX}} + \|2\mathbf{B}(\hat{\mathbf{E}} - \mathbf{E})(\hat{\mathbf{B}} - \mathbf{B})^{\top}\|_{\text{MAX}} \\
& \quad + \|(\hat{\mathbf{B}} - \mathbf{B})(\hat{\mathbf{E}} - \mathbf{E})(\hat{\mathbf{B}} - \mathbf{B})^{\top}\|_{\text{MAX}} + \|\hat{\Gamma}^{\text{S}} - \Gamma\|_{\text{MAX}}.
\end{aligned}$$

We need Lemmas 1 and 2, under the event of

$$\begin{aligned}\|\hat{\mathbf{E}} - \mathbf{E}\|_{\text{MAX}} &\leq C\sqrt{\Delta_n \log d}, \\ \max_{1 \leq j \leq r, 1 \leq l \leq d} \left| \sum_{i=1}^{\lfloor t/\Delta_n \rfloor} (\Delta_i^n F_j)(\Delta_i^n Z_l) \right| &\leq C\sqrt{\Delta_n \log d}, \\ \|\hat{\Gamma}^S - \Gamma\|_{\text{MAX}} &\leq \|\hat{\Gamma} - \Gamma\|_{\text{MAX}} \leq Cr\sqrt{\Delta_n \log d}, \\ \max_{i \leq d} \|\hat{\mathbf{b}}_i - \mathbf{b}_i\| &= O_p \left(\sqrt{\frac{r \log d}{T}} \right),\end{aligned}$$

for some C sufficiently large, and using $\|\mathbf{E}\| \leq r\|\mathbf{E}\|_{\text{MAX}} \leq rC$, $\max_{1 \leq k \leq d} \|\mathbf{b}_k\| \leq \sqrt{r}\|\mathbf{B}\|_{\text{MAX}}$, $\|\hat{\mathbf{B}} - \mathbf{B}\|_{\text{MAX}} \leq \sqrt{r}$, $\max_{1 \leq l \leq d} \|\hat{\mathbf{b}}_l - \mathbf{b}_l\|$, we have

$$\begin{aligned}\|2(\hat{\mathbf{B}} - \mathbf{B})\mathbf{E}\mathbf{B}^\top\|_{\text{MAX}} &= 2 \max_{1 \leq l, k \leq d} \|e_l^\top (\hat{\mathbf{B}} - \mathbf{B})\mathbf{E}\mathbf{B}^\top e_k\| \\ &\leq 2 \max_{1 \leq l \leq d} \|\hat{\mathbf{b}}_l - \mathbf{b}_l\| \|\mathbf{E}\| \max_{1 \leq k \leq d} \|\mathbf{b}_k\| \leq C'r^2 \sqrt{\frac{\log d}{T}}, \\ \|(\hat{\mathbf{B}} - \mathbf{B})\mathbf{E}(\hat{\mathbf{B}} - \mathbf{B})^\top\|_{\text{MAX}} &\leq \max_{1 \leq l \leq d} \|e_l^\top (\hat{\mathbf{B}} - \mathbf{B})\|^2 \|\mathbf{E}\| \\ &\leq C'r^2 \log d \sqrt{\frac{\Delta_n}{T}}, \\ \|2\mathbf{B}(\hat{\mathbf{E}} - \mathbf{E})(\hat{\mathbf{B}} - \mathbf{B})^\top\|_{\text{MAX}} &\leq 2r^2 \|\mathbf{B}\|_{\text{MAX}} \|\hat{\mathbf{E}} - \mathbf{E}\|_{\text{MAX}} \|\hat{\mathbf{B}} - \mathbf{B}\|_{\text{MAX}} \leq C'r^3 \log d \sqrt{\frac{\Delta_n}{T}}, \\ \|(\hat{\mathbf{B}} - \mathbf{B})(\hat{\mathbf{E}} - \mathbf{E})(\hat{\mathbf{B}} - \mathbf{B})^\top\|_{\text{MAX}} &\leq r^2 \|\hat{\mathbf{B}} - \mathbf{B}\|_{\text{MAX}}^2 \|\hat{\mathbf{E}} - \mathbf{E}\|_{\text{MAX}} \leq C'r^4 \Delta_n \log d \sqrt{\frac{\log d}{T}}.\end{aligned}$$

Finally, combining all these estimates, for some $C > 0$, we obtain

$$\|\hat{\Sigma}^{\text{MFACE}} - \Sigma\|_{\text{MAX}} \leq C \left(\sqrt{\frac{r^4 \log d}{T}} + \sqrt{r^2 \Delta_n \log d} \right).$$

Hence,

$$\|\hat{\Sigma}^{\text{MFACE}} - \Sigma\|_{\text{MAX}} = O_p \left(\sqrt{\frac{r^4 \log d}{T}} + \sqrt{r^2 \Delta_n \log d} \right).$$

Appendix 2. Proof of Lemma 3

(a) We note that

$$\begin{aligned}\|\hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}} - \mathbf{B}^\top \Gamma^{-1} \mathbf{B}\| &\leq 2\|(\hat{\mathbf{B}} - \mathbf{B})^\top \Gamma^{-1} \mathbf{B}\| + 2\|(\hat{\mathbf{B}} - \mathbf{B})^\top (\hat{\Gamma}^S)^{-1} - \Gamma^{-1}\| \mathbf{B}\| \\ &\quad + \|\mathbf{B}^\top ((\hat{\Gamma}^S)^{-1} - \Gamma^{-1}) \mathbf{B}\| + \|(\hat{\mathbf{B}} - \mathbf{B})^\top \Gamma^{-1} (\hat{\mathbf{B}} - \mathbf{B})\| \\ &\quad + \|(\hat{\mathbf{B}} - \mathbf{B})^\top (\hat{\Gamma}^S)^{-1} - \Gamma^{-1}\| (\hat{\mathbf{B}} - \mathbf{B})\|.\end{aligned}$$

This can be verified directly using Lemmas 1(b), 2(e), $\|\cdot\| \leq \|\cdot\|_F$ and $\|\mathbf{B}\| = O(\sqrt{d})$, which can be derived from Assumption 3(e).

(b) By Lemma 2(b) and $\|\cdot\| \leq \|\cdot\|_F$, we have with probability at least $1 - O(r^2 d^{-3})$,

$$\|\hat{\mathbf{E}} - \mathbf{E}\| \leq Cr\sqrt{\Delta_n \log d}.$$

By Lemma A.1 of Fan *et al.* (2011), we have

$$\begin{aligned}\mathbb{P}(\|\hat{\mathbf{E}}^{-1} - \mathbf{E}^{-1}\| < C'\|\hat{\mathbf{E}} - \mathbf{E}\|) &\geq \mathbb{P}(\|\hat{\mathbf{E}} - \mathbf{E}\| \\ &\leq C'r\sqrt{\Delta_n \log d}) \geq 1 - O(r^2 d^{-3}).\end{aligned}$$

Combining this with Lemma 3(a), and by the triangle inequality we have

$$\begin{aligned}\mathbb{P}(\|\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}} - \mathbf{B}^\top \Gamma^{-1} \mathbf{B} - \mathbf{E}^{-1}\| \\ < C'r\sqrt{\Delta_n \log d} + C'm_d r\sqrt{\Delta_n d^2 \log d}) > 1 - O(d^{-1}).\end{aligned}$$

Hence, the desired inequality follows from the fact that $d \times m_d \rightarrow \infty$.

(c) By their Lemma A.1 again, we have

$$\begin{aligned}\mathbb{P}(\lambda_{\min}(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}}) \geq Cd) \\ \geq \mathbb{P}(\|\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}} - \mathbf{B}^\top \Gamma^{-1} \mathbf{B} - \mathbf{E}^{-1}\| \leq Cd) \\ > 1 - O(d^{-1}),\end{aligned}$$

where we use the fact that $m_d r\sqrt{\Delta_n \log d} = o(1)$. This readily yields the desired inequality.

(d) By the triangle inequality and Lemma 1(b), we have, with probability approaching 1,

$$\begin{aligned}\|\hat{\mathbf{B}}\| &\leq \|\hat{\mathbf{B}}\|_F \leq \|\hat{\mathbf{B}} - \mathbf{B}\|_F + \|\mathbf{B}\|_F \leq C\sqrt{rd/T} + C\sqrt{d} \\ &\leq C'\sqrt{d}.\end{aligned}$$

On the other hand, with probability at least $1 - O(d^{-1})$, by Lemmas 2(f), 3(c), and the above inequality, there exists some C_0 sufficiently large, such that

$$\begin{aligned}\|\hat{\mathbf{B}}(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1}\| \\ \leq \|\hat{\mathbf{B}}\|^2 \cdot \|\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1}\| \cdot \|(\hat{\Gamma}^S)^{-1}\| \leq C_0.\end{aligned}$$

(e) This follows from the similar argument in Lemma 3(d) and that $\lambda_{\min}(\Gamma) \geq C$.

Appendix 3. Proof of Theorem 2

By the Sherman–Morrison–Woodbury formula, we have

$$\begin{aligned}\|(\hat{\Sigma}^{\text{MFACE}})^{-1} - \Sigma^{-1}\| &\leq \|(\hat{\Gamma}^S)^{-1} - \Gamma^{-1}\| \\ &\quad + \|((\hat{\Gamma}^S)^{-1} - \Gamma^{-1})\hat{\mathbf{B}}(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1}\| \\ &\quad + \|((\hat{\Gamma}^S)^{-1} - \Gamma^{-1})\hat{\mathbf{B}}(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \hat{\mathbf{B}}^\top \Gamma^{-1}\| \\ &\quad + \|\Gamma^{-1}(\hat{\mathbf{B}} - \mathbf{B})(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \hat{\mathbf{B}}^\top \Gamma^{-1}\| \\ &\quad + \|\Gamma^{-1}(\hat{\mathbf{B}} - \mathbf{B})(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \mathbf{B}^\top \Gamma^{-1}\| \\ &\quad + \|\Gamma^{-1} \mathbf{B}((\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \\ &\quad - (\mathbf{E}^{-1} + \mathbf{B}^\top (\Gamma^S)^{-1} \mathbf{B})^{-1}) \mathbf{B}^\top \Gamma^{-1}\| \\ &:= L_1 + L_2 + L_3 + L_4 + L_5 + L_6.\end{aligned}$$

We now bound each term above with probability no less than $1 - O(d^{-1})$.

First of all, by Lemma 2(e)

$$L_1 \leq Cm_d r \sqrt{\Delta_n \log d}.$$

To bound L_2 , we have, by Lemmas 2(e), 3(d),

$$\begin{aligned} L_2 &\leq \|(\hat{\Gamma}^S)^{-1} - \Gamma^{-1}\| \cdot \|\hat{\mathbf{B}}(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1}\| \\ &\leq Cm_d r \sqrt{\Delta_n \log d}. \end{aligned}$$

Similarly, L_3 can be bounded using Lemmas 2(e) and 3(e).

Next, for L_4 , we use Lemmas 1(b), 3(c), $\|\hat{\mathbf{B}}\| \leq \|\hat{\mathbf{B}}\|_F \leq C' \sqrt{d}$, $\|\cdot\| \leq \|\cdot\|_F$ and $\lambda_{\min}(\Gamma) \geq C$ in Assumption 2,

$$\begin{aligned} L_4 &\leq \|\Gamma^{-1}\|^2 \cdot \|\hat{\mathbf{B}} - \mathbf{B}\| \cdot \|\hat{\mathbf{B}}\| \cdot \|(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1}\| \\ &\leq C \sqrt{\Delta_n r \log d}. \end{aligned}$$

Similarly, using the fact that $\|\mathbf{B}\| \leq \|\mathbf{B}\|_F = O(\sqrt{d})$, we can establish the same bound for L_5 .

Finally, we have

$$\begin{aligned} L_6 &\leq \|\Gamma^{-1}\|^2 \cdot \|\mathbf{B}\|^2 \cdot \|(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1} \\ &\quad - (\mathbf{E}^{-1} + \mathbf{B}^\top (\Gamma^S)^{-1} \mathbf{B})^{-1}\| \\ &\leq \|\Gamma^{-1}\|^2 \cdot \|\mathbf{B}\|^2 \cdot \|(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}}) \\ &\quad - (\mathbf{E}^{-1} + \mathbf{B}^\top (\Gamma^S)^{-1} \mathbf{B})\| \\ &\quad \cdot \|(\hat{\mathbf{E}}^{-1} + \hat{\mathbf{B}}^\top (\hat{\Gamma}^S)^{-1} \hat{\mathbf{B}})^{-1}\| \cdot \|(\mathbf{E}^{-1} + \mathbf{B}^\top \Gamma^{-1} \mathbf{B})^{-1}\|. \end{aligned}$$

Note that for any vector v such that $\|v\| = 1$, by the definition of operator norm, we have

$$v^\top \mathbf{B}^\top \Gamma^{-1} \mathbf{B} v \geq \lambda_{\min}(\Gamma^{-1}) v^\top \mathbf{B}^\top \mathbf{B} v \geq \lambda_{\min}(\Gamma^{-1}) \lambda_{\min}(\mathbf{B}^\top \mathbf{B}).$$

It then follows that

$$\lambda_{\min}(\mathbf{B}^\top \Gamma^{-1} \mathbf{B}) \geq \lambda_{\min}(\Gamma^{-1}) \lambda_{\min}(\mathbf{B}^\top \mathbf{B}).$$

On the other hand, by Assumption 3(e), we have

$$\begin{aligned} d^{-1} v^\top \mathbf{B}^\top \mathbf{B} v &= v^\top \Omega v - v^\top (\Omega - d^{-1} \mathbf{B}^\top \mathbf{B}) v \\ &\geq \lambda_{\min}(\Omega) - \|d^{-1} \mathbf{B}^\top \mathbf{B} - \Omega\| > C, \end{aligned}$$

where C is a constant. Thus, $\lambda_{\min}(\mathbf{B}^\top \mathbf{B}) > Cd$. Therefore $\lambda_{\min}(\mathbf{B}^\top \Gamma^{-1} \mathbf{B}) > C'd$, where C' is a constant, following from the fact that $\lambda_{\max}(\Gamma) \leq C_0$. It then implies that $\lambda_{\min}(\mathbf{E}^{-1} + \mathbf{B}^\top \Gamma^{-1} \mathbf{B}) \geq \lambda_{\min}(\mathbf{B}^\top \Gamma^{-1} \mathbf{B}) > C'd$.

$$\|(\mathbf{E}^{-1} + \mathbf{B}^\top \Gamma^{-1} \mathbf{B})^{-1}\| = O_p(d^{-1}).$$

Using Lemmas 3(b) and 3(c), we further have

$$L_6 \leq Cm_d r \sqrt{\Delta_n \log d}.$$

Combining these estimates, we can obtain, for some $C > 0$,

$$\|(\hat{\Sigma}^{\text{MFACE}})^{-1} - \Sigma^{-1}\| \leq C \left(m_d r \sqrt{\Delta_n \log d} + \sqrt{\frac{r}{T}} \right).$$

Hence,

$$\|(\hat{\Sigma}^{\text{MFACE}})^{-1} - \Sigma^{-1}\| = O_p \left(m_d r \sqrt{\Delta_n \log d} + \sqrt{\frac{r}{T}} \right).$$